

State-Level Economic Similarity and Structural Stability: An Unsupervised Learning Analysis of U.S. States Before, During, and After the 2020–2022 Economic Shock

An Experimental Framework for Examining State-Level Economic Similarity and Structural Stability

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Acknowledgment

I developed this project with assistance from OpenAI Codex using GPT-5.6 Sol, which supported my analytical work and learning process.

I used the tool for technical discussions, Python code debugging, methodological clarification, workflow development, and interpretation of machine learning concepts and results. It also supported my work on data preparation, exploratory data analysis, PCA, clustering, and the design of subsequent analytical procedures.

I independently conducted the analysis, executed and validated the Python code, reviewed the analytical outputs and recommendations, and made all final decisions regarding the methodology, interpretations, conclusions, and report content.

I acknowledge this assistance as AI-supported analytical and educational guidance. The tool did not serve as an independent author or replace my judgment and validation.

Project Objectives

Research Questions

This analytical portfolio project examines U.S. state-level economic similarity and structural stability through four research questions:

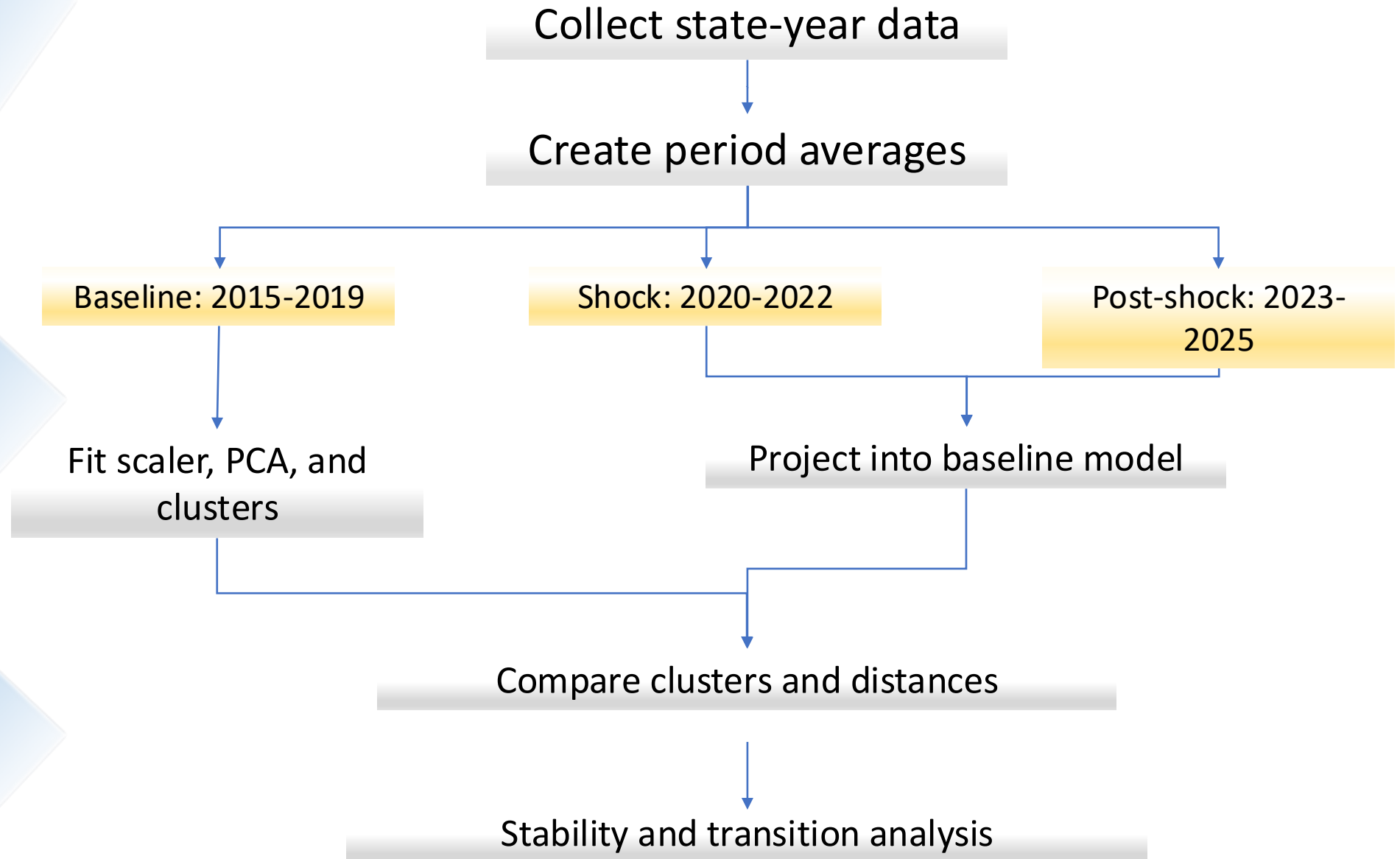
1. Which U.S. states had similar economic and industry-employment structures during the 2015-2019 baseline period?
2. How did structural similarities among states change during the 2020-2022 shock period and the 2023- 2024 post-shock period?
3. Which states remained structurally stable, temporarily changed, or experienced a possible structural shift across the three periods?
4. Which states were the closest economic peers of Illinois during each research period, and did Illinois's peer group change over time?

Analytical Workflow

Data
Preparation

Processes of
EDA, PCA,
Clustering

Unsupervised
Machine Learning
Modeling



Analytical Scope

Geographic and temporal scope

Element	Scope
Geography	50 U.S. states
Excluded	District of Columbia, Puerto Rico and national totals
Baseline	2015–2019
Shock	2020–2022
Post-shock	2023–2024
Full period	2015–2024
Unit used for clustering	One averaged observation per state per period

Data Collection

Table Codes	Purposes	Sources
SAGDP9N	Real GDP by state and industry	U.S. Bureau of Economic Analysis (BEA)
SAINC1	State personal income summary	
SAINC30	State economic profile and earnings measures	
QCEW (Quarterly Versus Employment and Wages)	Employment by industry	U.S. Bureau of Labor Statistics (BLS)

BEA Collection Summary

	Table	Rows	Columns	First_Column	Last_Column
0	SAGDP9	5524	37	GeoFIPS	2025
1	SAINC1	184	105	GeoFIPS	2025
2	SAINC30	2284	75	GeoFIPS	2024

BLS-QCEW Collection Summary

Industry coverage

	Industry	Non_Missing	Missing	Years_Available	Coverage_Percent
1	Mining	550	11	11	98.039216
0	Agriculture	559	2	11	99.643494
2	Utilities	561	0	11	100.000000
3	Construction	561	0	11	100.000000
4	Manufacturing	561	0	11	100.000000
5	Wholesale_Trade	561	0	11	100.000000
6	Retail_Trade	561	0	11	100.000000
7	Transportation_Warehousing	561	0	11	100.000000
8	Information	561	0	11	100.000000
9	Finance_Insurance	561	0	11	100.000000
10	Real_Estate	561	0	11	100.000000
11	Professional_Technical	561	0	11	100.000000
12	Management_of_Companies	561	0	11	100.000000
13	Administrative_Support	561	0	11	100.000000
14	Educational_Services	561	0	11	100.000000
15	Health_Care	561	0	11	100.000000
16	Arts_Entertainment	561	0	11	100.000000
17	Accommodation_Food	561	0	11	100.000000
18	Other_Services	561	0	11	100.000000
19	Public_Administration	561	0	11	100.000000

Employment coverage

	year	Rows	State_Areas	Missing
0	2015	53	53	0
1	2016	53	53	0
2	2017	53	53	0
3	2018	53	53	0
4	2019	53	53	0
5	2020	53	53	0
6	2021	53	53	0
7	2022	53	53	0
8	2023	53	53	0
9	2024	53	53	0
10	2025	53	53	0

Data Preparation

Step 1. Merge datasets

Merging summary

	Dataset	Rows	Columns	Missing_Cells	Duplicate_Rows
0	SAGDP9	5524	37	144	0
1	SAINC1	184	105	416	0
2	SAINC30	2284	75	296	0
3	BLS_Industry	561	22	13	0

Geographic cleaning summary

	Dataset	Rows_Retained	States_Retained
0	SAGDP9	4600	50
1	SAINC1	150	50
2	SAINC30	1900	50
3	BLS Industry	550	50

- The geographic identifiers were standardized across the BEA and BLS datasets.
- National totals, Washington, D.C., and other non-state geographic areas were excluded.
- All four datasets now contain the same 50-state population.
- The BLS dataset contains one row per state-year.

BEA cleaning summary

	Dataset	Rows	Unique_States
0	SAGDP9	4600	50
1	SAINC1	150	50
2	SAINC30	1900	50

BLS cleaning summary

	year	Rows	Unique_States
0	2015	50	50
1	2016	50	50
2	2017	50	50
3	2018	50	50
4	2019	50	50
5	2020	50	50
6	2021	50	50
7	2022	50	50
8	2023	50	50
9	2024	50	50
10	2025	50	50

Step 2. Statistics in BEA Dataset

BEA Summary

	Feature	Rows	States	First_Year	Last_Year	Missing
0	Real GDP	500	50	2015	2024	0
1	Population	500	50	2015	2024	0
2	Personal income per capita	500	50	2015	2024	0

	real_gdp_per_capita	personal_income_per_capita
count	500.0	500.0
mean	59077.39777	56974.854
std	11028.352528	11557.946947
min	36650.485762	35100.0
25%	50987.285561	48183.75
50%	58152.203365	55769.5
75%	65564.505796	64203.25
max	91996.437853	95083.0

Step 3. Economy-profile selection from SAINC30 (BEA)

Line code	Measure	Use
240	Total employment—number of jobs	Calculate real output per job
300	Average wages and salaries	Wage-level feature



	Feature	LineCode	Rows	States	First_Year	Last_Year	Missing
0	Total employment	240	500	50	2015	2024	0
1	Average wages and salaries	300	500	50	2015	2024	0

Step 4. Economic features summary

	count	mean	std	min	25%	50%	75%	max
real_gdp_per_capita	500.0	59077.39777	11028.352528	36650.485762	50987.285561	58152.203365	65564.505796	91996.437853
personal_income_per_capita	500.0	56974.854	11557.946947	35100.0	48183.75	55769.5	64203.25	95083.0
total_employment_jobs	500.0	4032169.786	4526384.53978	398144.0	1067533.75	2642780.5	4869518.25	25583169.0
average_wages_salaries	500.0	58020.652	11108.127296	38579.0	49488.25	56505.0	64650.5	94790.0
real_output_per_job	500.0	95508.086214	14434.870112	68733.424874	86103.749202	93046.569412	101096.642886	143192.315849

Comment:

- Total employment and average wages were selected from SAINC30 using verified line codes 240 and 300.
- Total employment was combined with real GDP to calculate real output per job. This measure serves as a labor-productivity proxy, but it does not account for differences in hours worked or multiple-job holders.
- Average wages and personal income are currently nominal dollar measures.
- The updated BEA dataset contains one observation for every state-year from 2015 through 2024.

Step 5. Prepare BLS industry data for 2015-2024

Total Industries: 20	
Agriculture	Real Estate
Mining	Professional Technical
Utilities	Management of Companies
Construction	Administrative Support
Manufacturing	Educational Services
Wholesale Trade	Health Care
Retail Trade	Arts Entertainment
Transportation Warehousing	Accommodation Food
Information	Other Services
Finance Insurance	Public Administration

Split into four industry shares



Examine shares quality

	Feature	Missing	Minimum	Maximum	Below_Zero	Above_100
0	manufacturing_share	0	0.000000	17.754177	0	0
1	professional_business_share	0	6.637846	20.460170	0	0
2	healthcare_share	0	9.618643	21.305872	0	0
3	natural_resources_agriculture_share	0	0.000000	9.360484	0	0

Step 6. Merge BEA and BLS Economic Features

1. Merge Summary

	Merge_Status	Rows
0	both	500
1	left_only	0
2	right_only	0

2. Combined Structure information

	Measure	Result
0	Rows	500
1	Columns	16
2	States	50
3	Years	10
4	First year	2015
5	Last year	2024
6	Duplicate state-years	0

3. Check feature quality

	Feature	Missing	Coverage_Percent	Status
0	real_gdp_per_capita	0	100.0	High
1	personal_income_per_capita	0	100.0	High
2	real_output_per_job	0	100.0	High
3	average_wages_salaries	0	100.0	High
4	manufacturing_share	0	100.0	High
5	professional_business_share	0	100.0	High
6	healthcare_share	0	100.0	High
7	natural_resources_agriculture_share	0	100.0	High

4. Merged features statistics

	count	mean	std	min	25%	50%	75%	max
real_gdp_per_capita	500.0	59077.39777	11028.352528	36650.485762	50987.285561	58152.203365	65564.505796	91996.437853
personal_income_per_capita	500.0	56974.854	11557.946947	35100.0	48183.75	55769.5	64203.25	95083.0
real_output_per_job	500.0	95508.086214	14434.870112	68733.424874	86103.749202	93046.569412	101096.642886	143192.315849
average_wages_salaries	500.0	58020.652	11108.127296	38579.0	49488.25	56505.0	64650.5	94790.0
manufacturing_share	500.0	8.847927	3.523305	0.0	6.104998	9.072297	11.18266	17.754177
professional_business_share	500.0	13.200289	2.595176	6.637846	11.707428	13.463461	14.738785	20.46017
healthcare_share	500.0	15.205974	1.970203	9.618643	13.966636	14.877686	16.725712	21.305872
natural_resources_agriculture_share	500.0	1.566749	1.502633	0.0	0.5761	1.087786	1.983248	9.360484

Step 7. Adjust nominal variables for inflation

Create a CPI-U table

	year	cpi_u
0	2015	237.017
1	2016	240.007
2	2017	245.120
3	2018	251.107
4	2019	255.657
5	2020	258.811
6	2021	270.970
7	2022	292.655
8	2023	304.702
9	2024	313.689

Base year

$$\text{Real value in 2017 dollars} = \text{Nominal value} \times \frac{\text{CPI}_{2017}}{\text{CPI}_{\text{year}}}$$

- This project uses 2017 as the inflation-adjustment base year because the real GDP variables are reported in chained 2017 dollars.
- Nominal income and wage values will therefore be converted into consent 2017 dollars.

Step 8. The summary of the eight features

Interpretation

Dimension	Available features
Economic capacity	GDP and income per capita
Productivity	Output per job
Labor compensation	Real wages
Industry employment structure	Four employment shares

Final quality summary

	Feature	Non_Missing	Missing	Infinite	Minimum	Maximum	Coverage_Percent
0	real_gdp_per_capita	500	0	0	36650.485762	91996.437853	100.0
1	real_personal_income_per_capita_2017	500	0	0	36299.978483	76105.974241	100.0
2	real_output_per_job	500	0	0	68733.424874	143192.315849	100.0
3	real_average_wages_salaries_2017	500	0	0	39897.916521	78771.836292	100.0
4	manufacturing_share	500	0	0	1.038760	17.754177	100.0
5	professional_business_share	500	0	0	6.637846	20.460170	100.0
6	healthcare_share	500	0	0	9.618643	21.305872	100.0
7	natural_resources_agriculture_share	500	0	0	0.039280	9.360484	100.0

Step 9. Create the three research periods

	year	period	
0	2015	Baseline_2015_2019	Baseline: 2015- 2019
1	2016	Baseline_2015_2019	
2	2017	Baseline_2015_2019	
3	2018	Baseline_2015_2019	
4	2019	Baseline_2015_2019	
5	2020	Shock_2020_2022	Shock: 2020- 2022
6	2021	Shock_2020_2022	
7	2022	Shock_2020_2022	
8	2023	Post_Shock_2023_2024	Post-shock: 2023-2024
9	2024	Post_Shock_2023_2024	

Step 10. Data Preparation conclusion

- The final project uses **eight** fully prepared structural features covering **economic capacity, productivity, compensation, and industry-employment composition**.
- Four annual growth indicators were constructed from the existing data to support the analysis of the 2020–2022 shock. The structural variables will be used for PCA and clustering, while the growth variables will help interpret economic changes within and between clusters.
- The final panel contains **500** observations representing **50 states from 2015 through 2024**. It is divided into **baseline, shock, and post-shock** periods.

Unsupervised Machine Learning Modeling

Step 1. Define features groups

Structural Features
real_gdp_per_capita
real_personal_income_per_capita_2017
real_output_per_job
real_average_wages_salaries_2017
manufacturing_share
professional_business_share
healthcare_share
natural_resources_agriculture_share

Growth Features
real_gdp_per_capita_growth
real_income_per_capita_growth
total_employment_growth
real_average_wage_growth

Period Order
Baseline_2015_2019
Shock_2020_2022
Post_Shock_2023_2024

Step 2. Exploratory Data Analysis

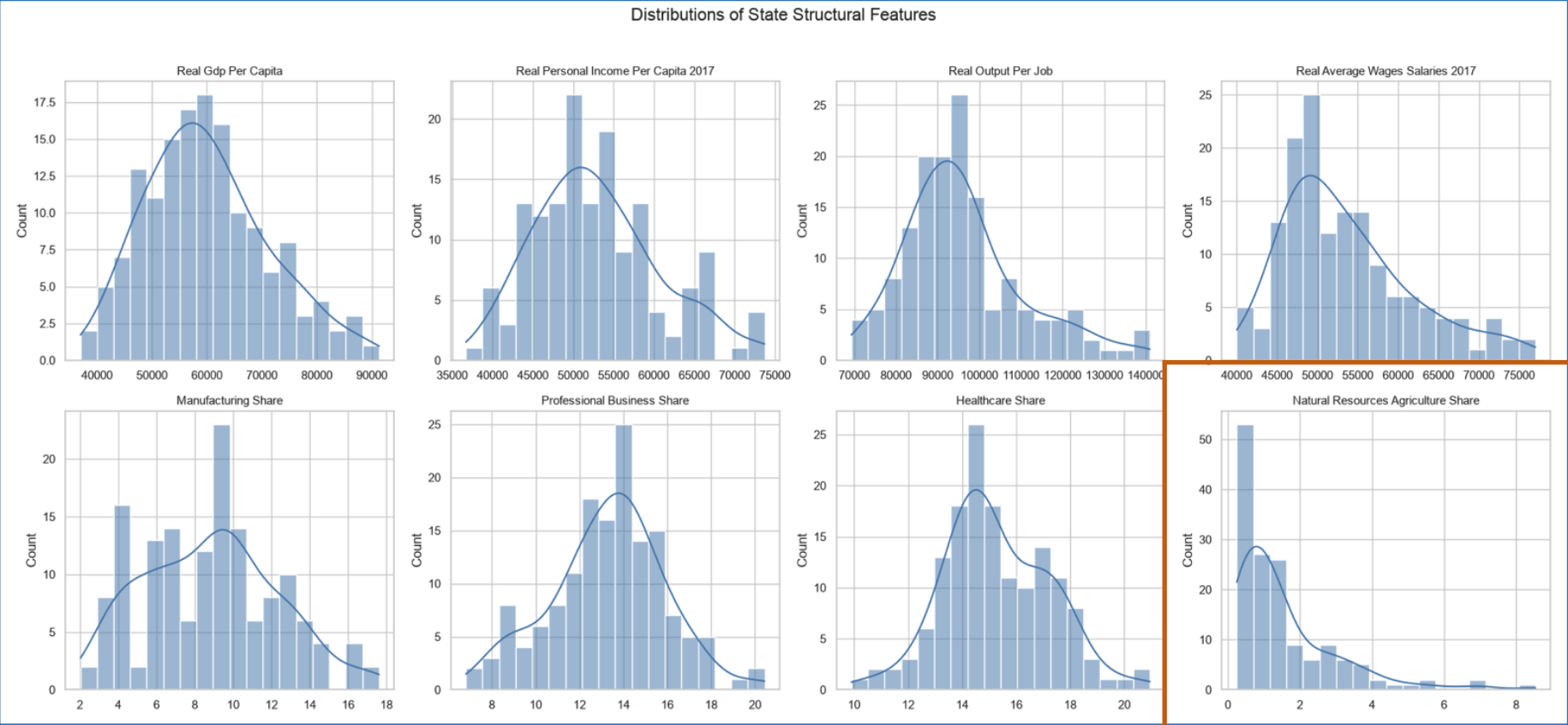
Statistics summary- Structural Features

	count	mean	std	min	25%	50%	75%	max	skewness	range
real_gdp_per_capita	150.0000	59,983.2897	11,135.8590	36,987.5761	52,076.5492	58,646.0948	66,618.4936	91,316.5067	0.4942	54,328.9306
real_personal_income_per_capita_2017	150.0000	52,981.5631	7,799.6713	36,706.3789	47,751.3675	52,172.4733	57,886.7103	73,710.0400	0.5375	37,003.6611
real_output_per_job	150.0000	96,264.8447	14,529.4047	69,295.3655	86,656.5502	94,009.9661	101,353.6797	140,945.5814	0.9277	71,650.2159
real_average_wages_salaries_2017	150.0000	53,647.7229	8,035.6776	40,017.7027	47,992.6643	51,984.2323	57,886.8322	76,999.9313	0.9295	36,982.2285
manufacturing_share	150.0000	8.8202	3.4985	2.0320	6.1020	9.0738	11.1467	17.6143	0.2572	15.5823
professional_business_share	150.0000	13.2766	2.5955	6.8161	11.8503	13.5640	14.8088	20.4306	-0.1119	13.6145
healthcare_share	150.0000	15.2876	1.9703	9.8995	14.0012	14.9980	16.7282	20.9485	0.2100	11.0490
natural_resources_agriculture_share	150.0000	1.5472	1.4513	0.2487	0.5520	1.0704	1.9543	8.5348	2.0521	8.2862

Period feature means

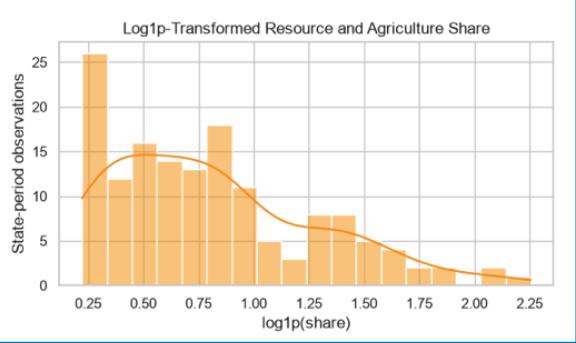
	period	Baseline_2015_2019	Shock_2020_2022	Post_Shock_2023_2024
real_gdp_per_capita		57,106.7794	59,553.9012	63,289.1885
real_personal_income_per_capita_2017		49,442.3725	54,646.8401	54,855.4768
real_output_per_job		93,455.9831	97,123.8443	98,214.7068
real_average_wages_salaries_2017		51,482.8634	55,238.6679	54,221.6374
manufacturing_share		8.9146	8.8947	8.6513
professional_business_share		12.9523	13.4862	13.3913
healthcare_share		14.9881	15.3696	15.5052
natural_resources_agriculture_share		1.6529	1.5031	1.4858

Step 3. Visualization

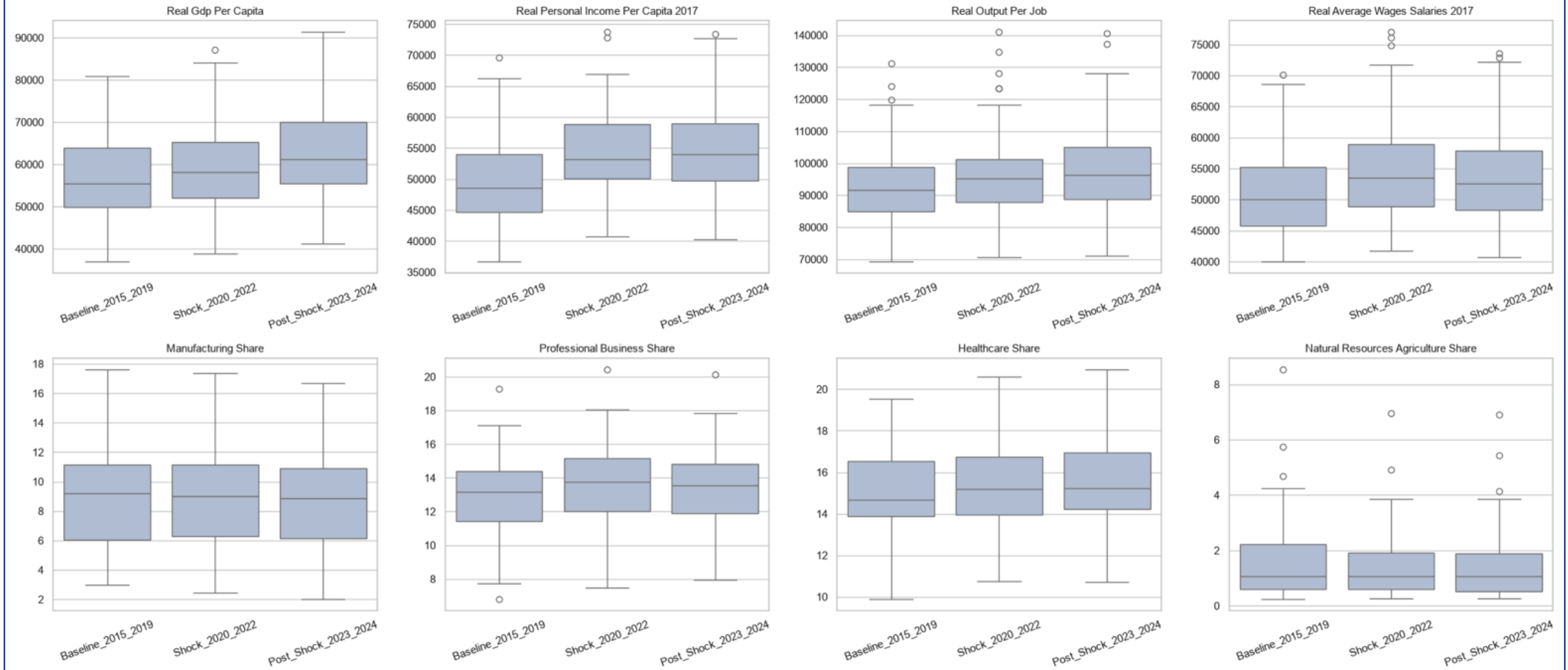


Improved skewness

Version		Feature	Skewness	Absolute_Skewness
0	Original	natural_resources_agriculture_share	2.0521	2.0521
1	Log1p transformed	log1p_natural_resources_agriculture_share	0.8309	0.8309



Structural Features Before, During and After the Shock

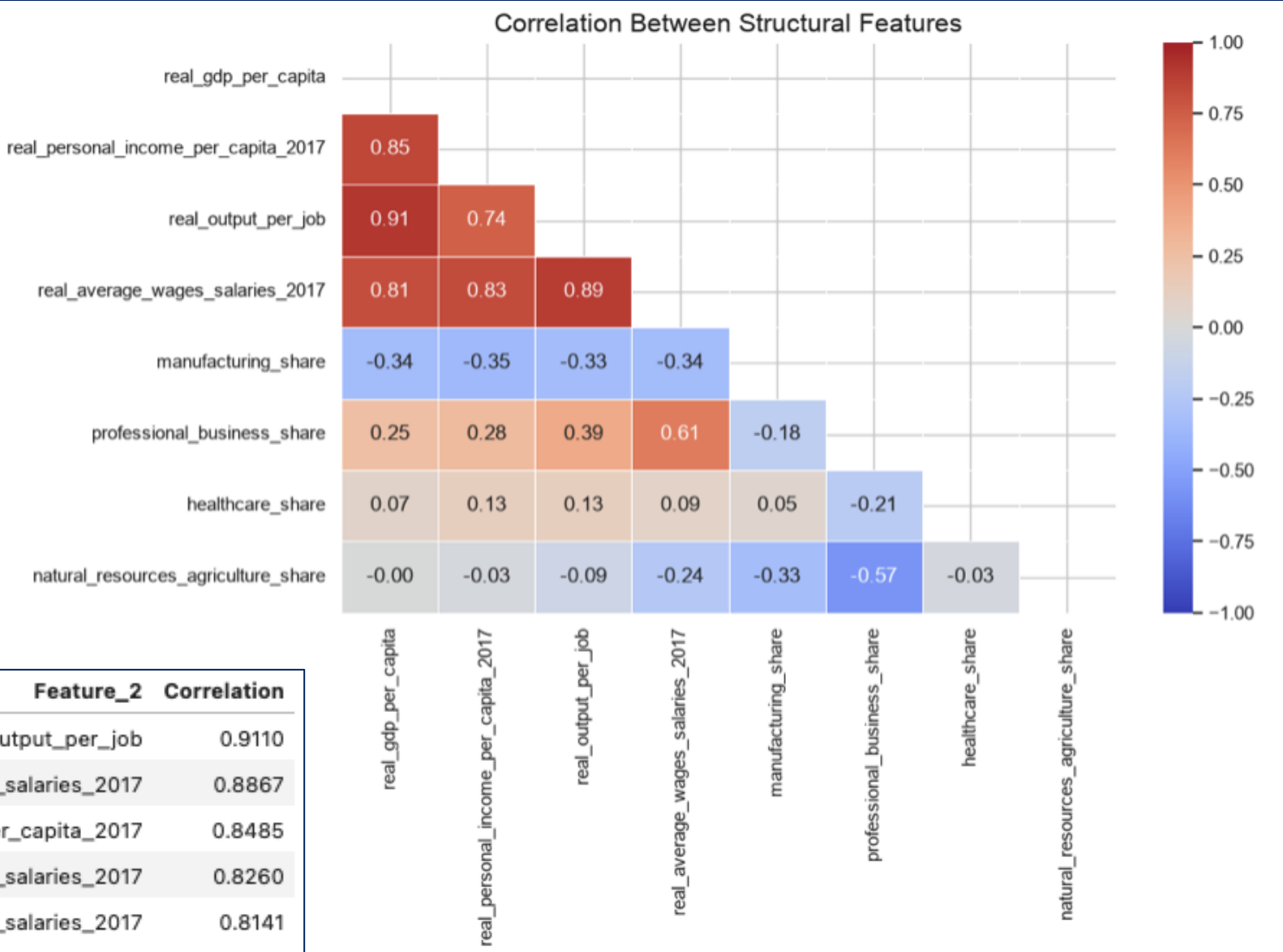


Interpretation:

- The boxplots compare the cross-state distributions of each structural feature across the baseline, shock, and post-shock periods.
- A shift in a period's median represents a broad change across states.
- Changes in the box width indicate changes in interstate dispersion.

Interpretation:

- Strong correlations are expected among GDP per capita, income, wages, and output per job because these variables represent related dimensions of economic capacity.
- In this project, the correlation matrix is used to identify overlapping information and to support the PCA interpretation.



Strong correlations

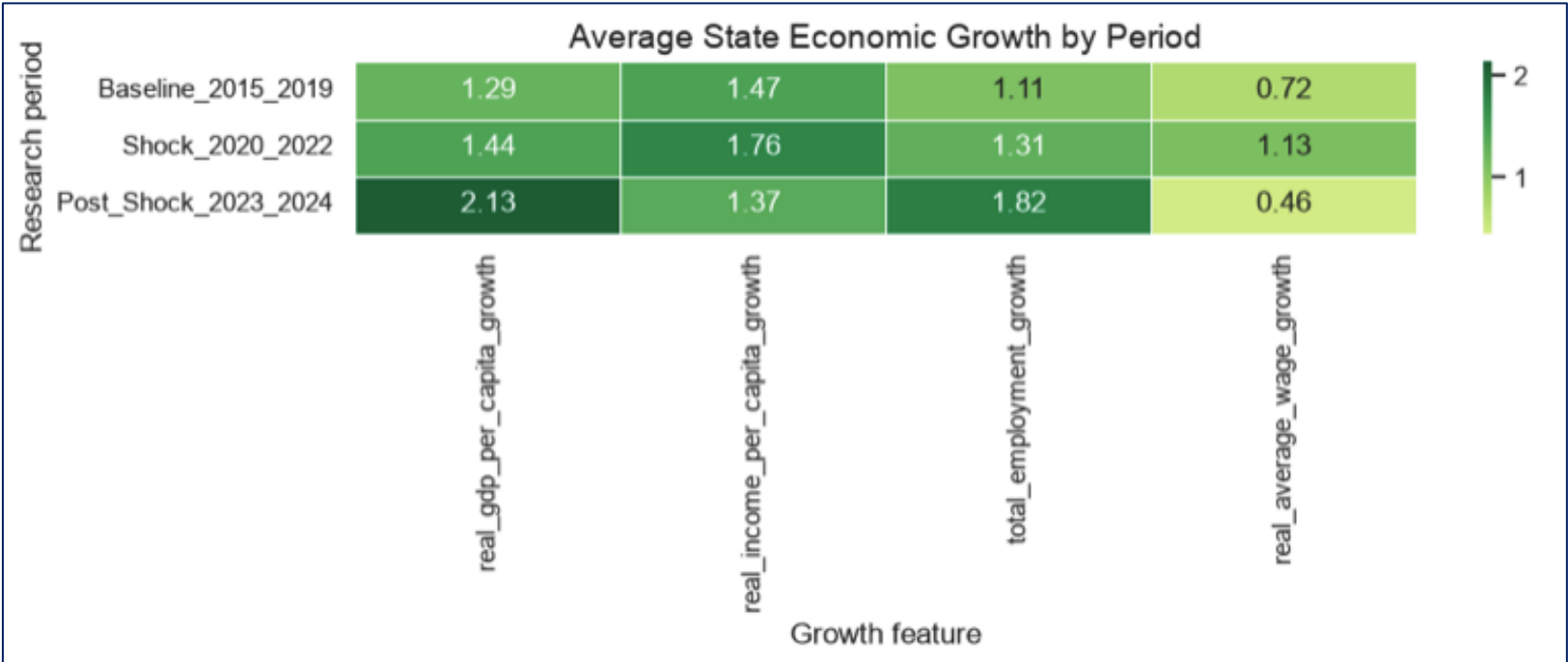
	Feature_1	Feature_2	Correlation
1	real_gdp_per_capita	real_output_per_job	0.9110
4	real_output_per_job	real_average_wages_salaries_2017	0.8867
0	real_gdp_per_capita	real_personal_income_per_capita_2017	0.8485
3	real_personal_income_per_capita_2017	real_average_wages_salaries_2017	0.8260
2	real_gdp_per_capita	real_average_wages_salaries_2017	0.8141

Statistics summary- **Growth Features**

period	real_gdp_per_capita_growth			real_income_per_capita_growth			total_employment_growth			real_average_wage_growth		
	mean	median	std	mean	median	std	mean	median	std	mean	median	std
Baseline_2015_2019	1.2924	1.2188	1.0238	1.4742	1.4050	0.7693	1.1096	0.8978	0.9453	0.7201	0.7021	0.5491
Shock_2020_2022	1.4446	1.5933	1.1393	1.7600	1.6896	0.9258	1.3148	1.0823	1.0194	1.1294	1.0238	0.8159
Post_Shock_2023_2024	2.1286	1.9932	1.2023	1.3660	1.5435	0.9036	1.8168	1.7264	0.3785	0.4588	0.4892	0.6168

Interpretation:

- The growth indicators provide supporting evidence about economic movement during each period.
- Negative or weaker average growth during 202-2022 would indicate the economic disruption associated with shock period.
- Post-shock growth helps show whether states recovered at similar or different rates.
- These results describe economic patterns but do not establish that the pandemic caused every observed change.



Step 4. Standardize structural modeling features

	Scaled_Mean	Scaled_Standard_Deviation	Scaled_Minimum	Scaled_Maximum
real_gdp_per_capita	0.0000	1.0000	-2.0719	2.8231
real_personal_income_per_capita_2017	-0.0000	1.0000	-2.0936	2.6665
real_output_per_job	-0.0000	1.0000	-1.8624	3.0855
real_average_wages_salaries_2017	-0.0000	1.0000	-1.7019	2.9158
manufacturing_share	-0.0000	1.0000	-1.9468	2.5221
professional_business_share	-0.0000	1.0000	-2.4975	2.7655
healthcare_share	0.0000	1.0000	-2.7439	2.8828
log1p_natural_resources_agriculture_share	-0.0000	1.0000	-1.2857	3.1192

	Original_Feature	Modeling_Feature	Transformation	Standardized
0	real_gdp_per_capita	real_gdp_per_capita	None	True
1	real_personal_income_per_capita_2017	real_personal_income_per_capita_2017	None	True
2	real_output_per_job	real_output_per_job	None	True
3	real_average_wages_salaries_2017	real_average_wages_salaries_2017	None	True
4	manufacturing_share	manufacturing_share	None	True
5	professional_business_share	professional_business_share	None	True
6	healthcare_share	healthcare_share	None	True
7	natural_resources_agriculture_share	log1p_natural_resources_agriculture_share	log1p	True

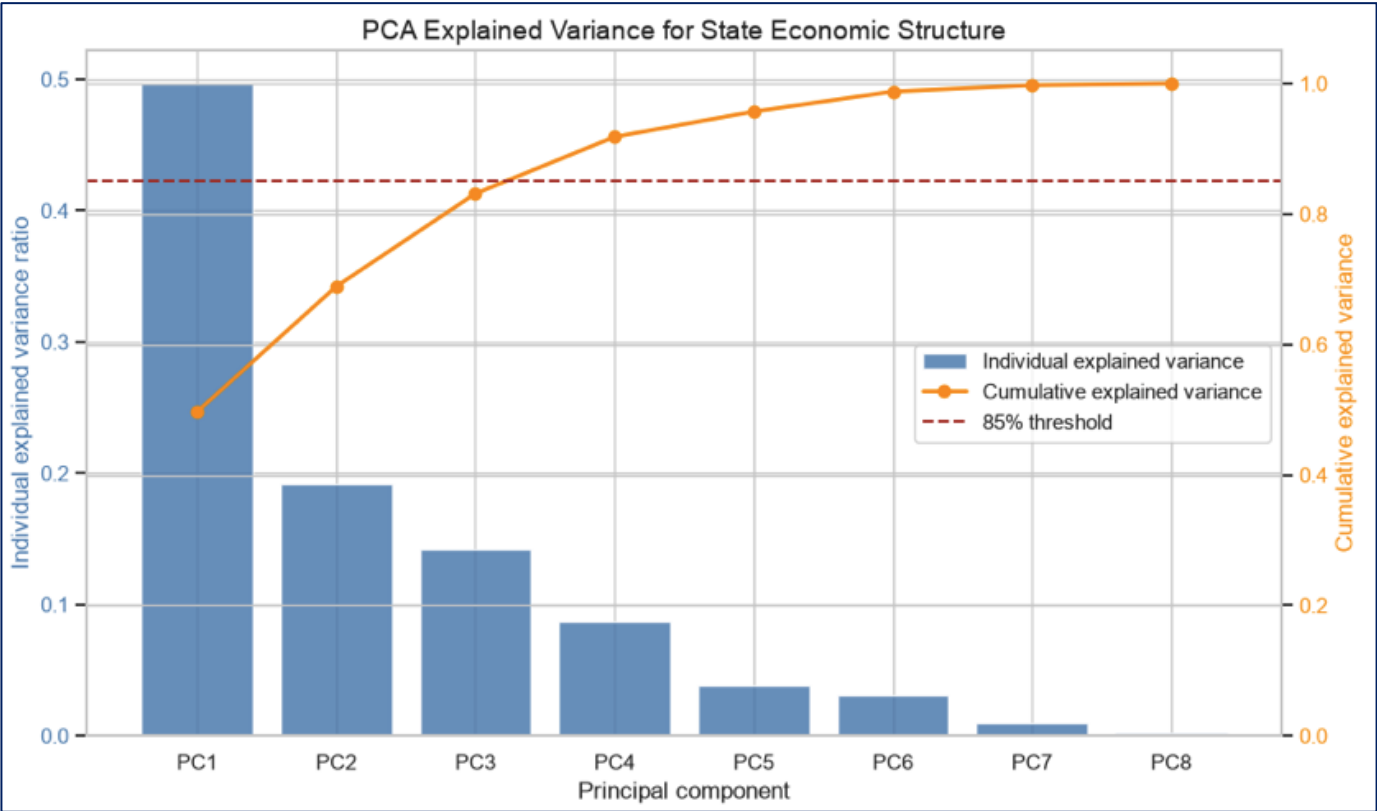
	Feature	Pre_Scaling_Mean	Pre_Scaling_Standard_Deviation
0	real_gdp_per_capita	59,983.2897	11,098.6774
1	real_personal_income_per_capita_2017	52,981.5631	7,773.6289
2	real_output_per_job	96,264.8447	14,480.8923
3	real_average_wages_salaries_2017	53,647.7229	8,008.8472
4	manufacturing_share	8.8202	3.4868
5	professional_business_share	13.2766	2.5868
6	healthcare_share	15.2876	1.9637
7	log1p_natural_resources_agriculture_share	0.8154	0.4615

Interpretation:

- All 8 analytical features will be standardized using `StandardScaler`.
- A single scaler will be fitted across the complete analytical sample so that differences between the baseline, shock, post-shock periods are preserve.

Step 5. Principal Component Analysis (PCA)

Total explained variance



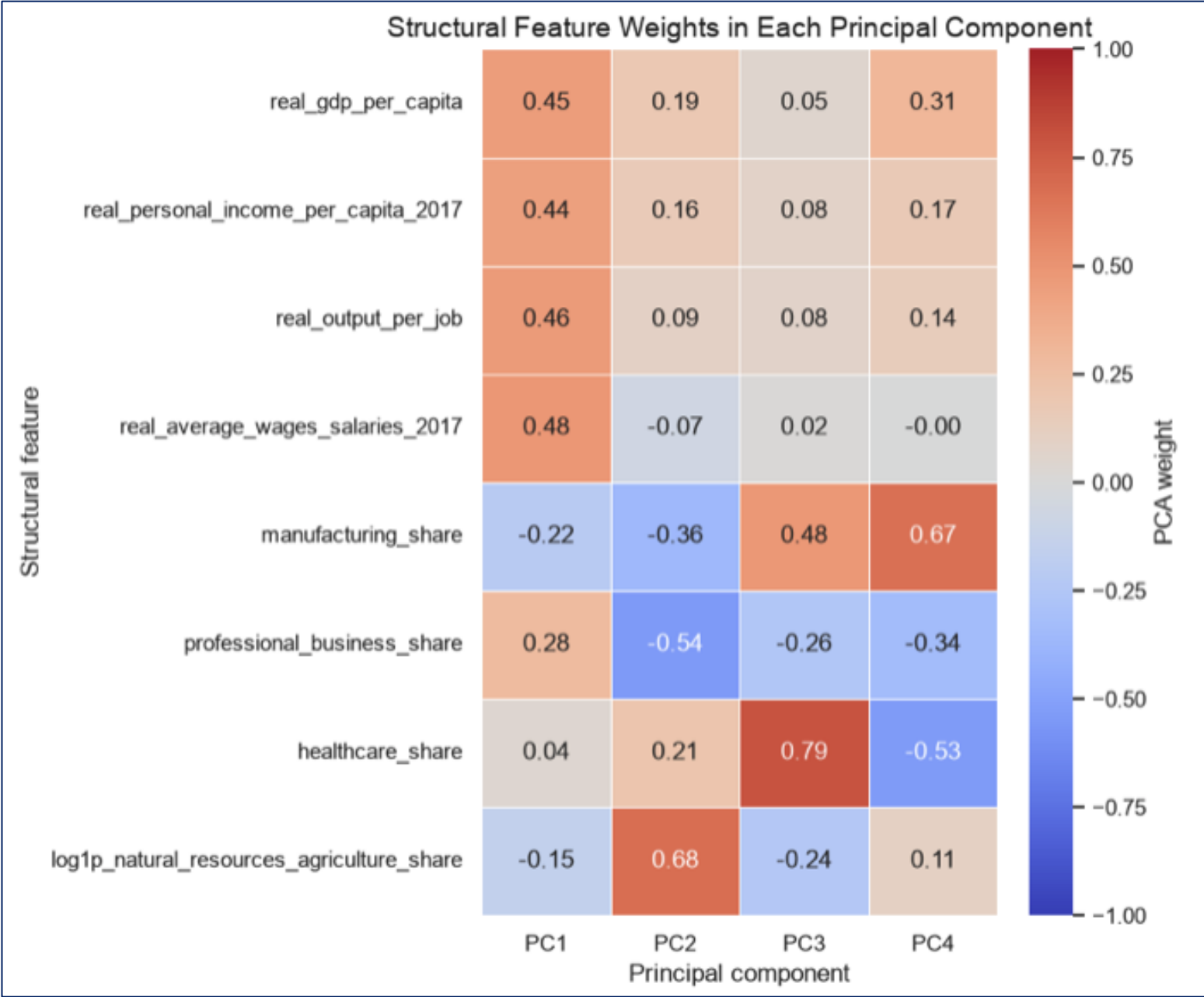
Interpretation:

- The bars show the information explained by each individual component.
- The orange line shows cumulative information retained.
- The dashed line marks the **85%** selection threshold.
- An elbow indicates where additional components begin providing relatively little information.

Component	Variance	Cumulative	Suggested interpretation
PC1	49.68%	49.68%	Economic capacity and productivity
PC2	19.20%	68.88%	Resource orientation versus business/industrial structure
PC3	14.24%	83.12%	Healthcare and manufacturing concentration
PC4	8.70%	91.82%	Manufacturing versus healthcare/service orientation

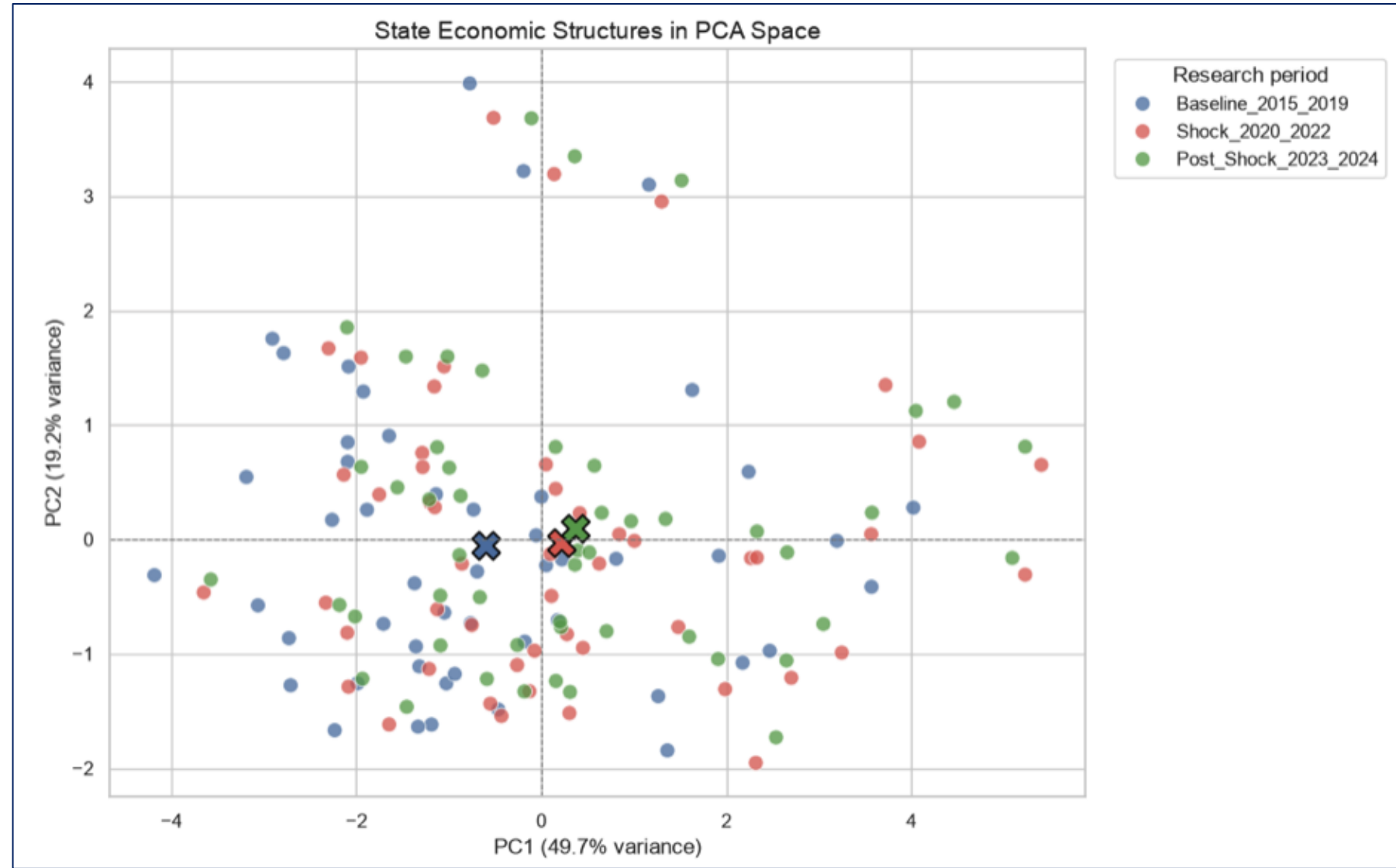
Interpretation:

- A large positive or negative value means the feature strongly defines that component.
- The sign shows direction, but PCA signs are mathematically reversible.



Interpretation:

- The PCA score plot shows substantial overlap among the baseline, shock, and post-shock periods.
- The period centroids shift slightly toward higher PC1 values, but interstate structural differences remain much larger than the average differences between periods.



Step 6. Clustering and K-Means Models

Clustering inputs

PCA Components	4
Components used	PC1, PC2, PC3, PC4
Observations	150

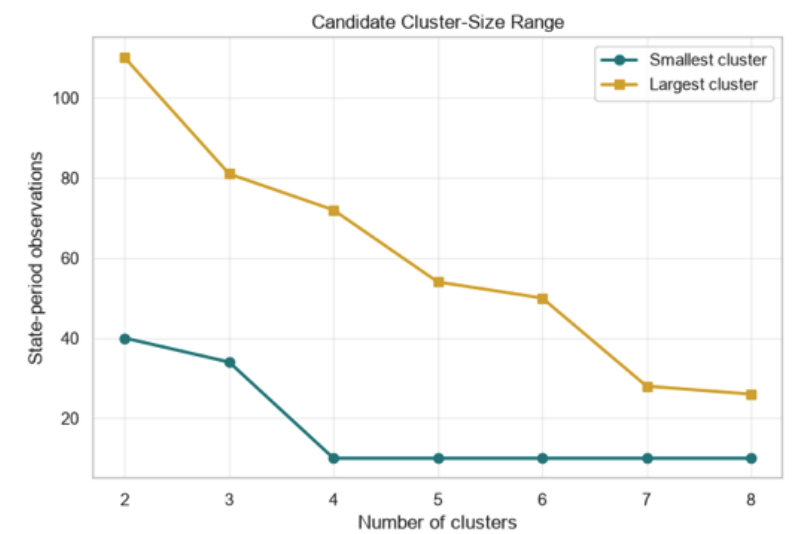
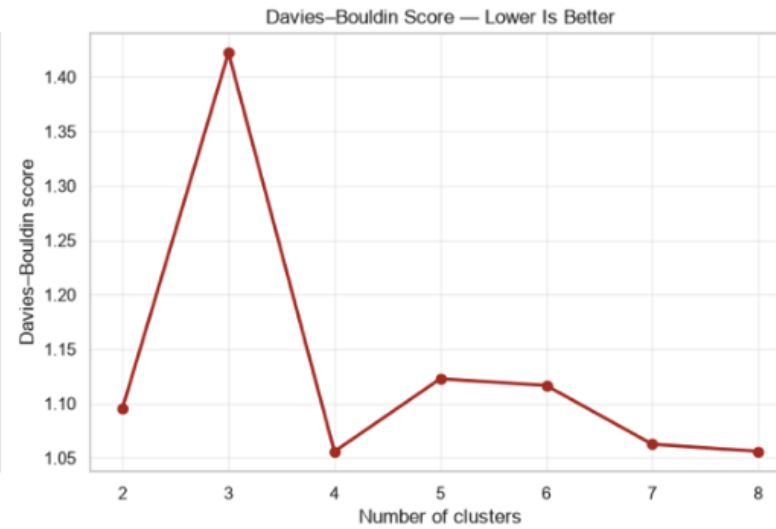
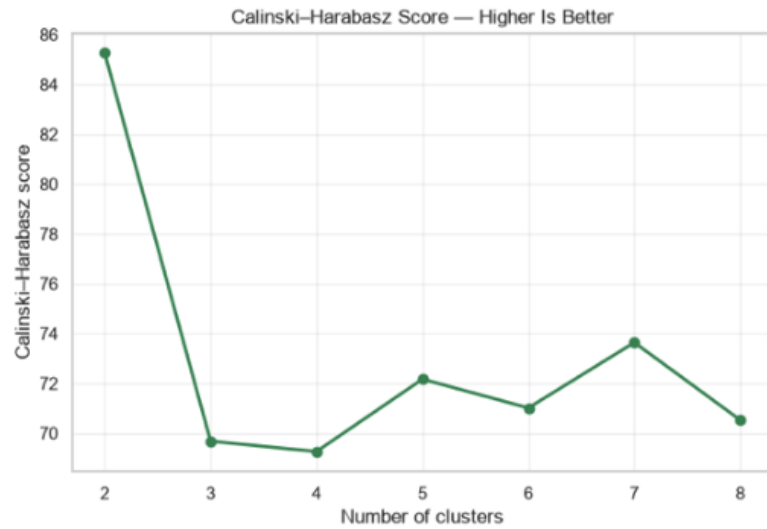
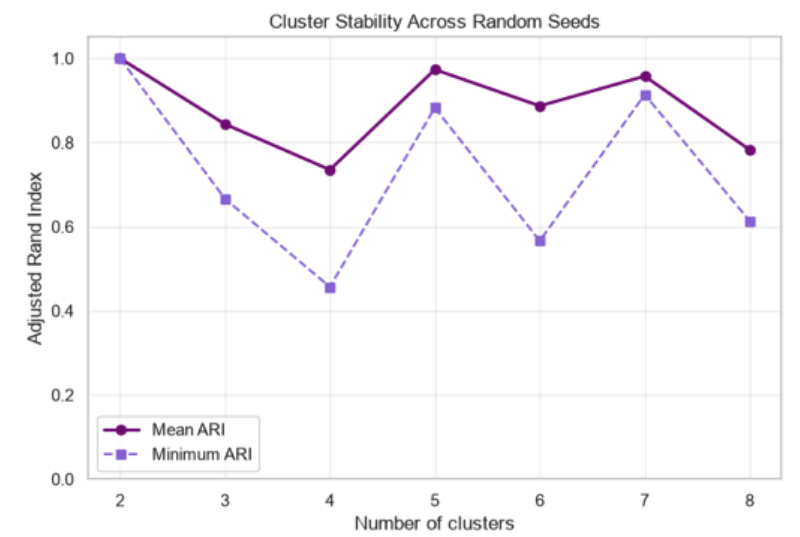
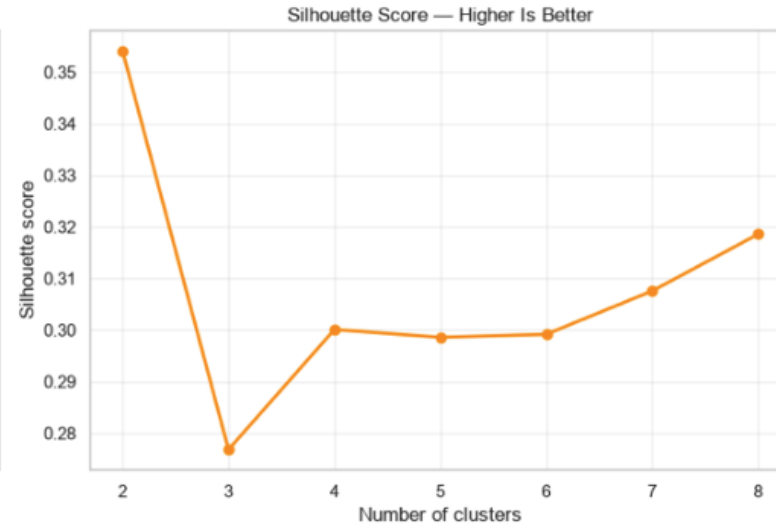
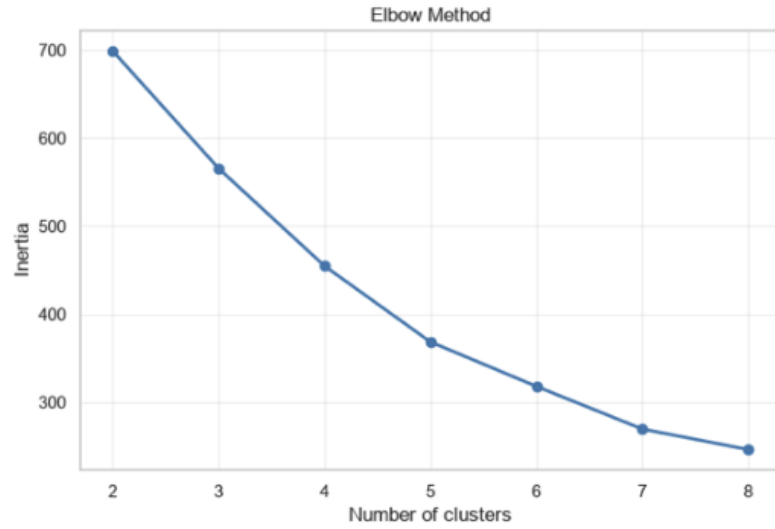
Evaluate candidate K-Means model

	Number_of_Clusters	Inertia	Silhouette_Score	Calinski_Harabasz_Score	Davies_Bouldin_Score	Smallest_Cluster	Largest_Cluster	Smallest_Cluster_Percent
0	2	699.0917	0.3541	85.2646	1.0960	40	110	26.6667
1	3	565.6310	0.2768	69.6777	1.4221	34	81	22.6667
2	4	454.7071	0.3001	69.2625	1.0559	10	72	6.6667
3	5	368.4097	0.2986	72.1672	1.1228	10	54	6.6667
4	6	317.9277	0.2992	71.0126	1.1165	10	50	6.6667
5	7	269.4150	0.3076	73.6397	1.0626	10	28	6.6667
6	8	246.1355	0.3186	70.5250	1.0561	10	26	6.6667

Criteria Interpretation

Criterion	Preferred direction	Meaning
Inertia	Lower	Observations are closer to their assigned centroids
Silhouette	Higher	Clusters are compact and well separated
Calinski–Harabasz	Higher	Between-cluster separation is large relative to within-cluster variation
Davies–Bouldin	Lower	Clusters have less overlap
Smallest cluster	Review	Detects solutions containing very small clusters

K-Means Cluster-Number Evaluation



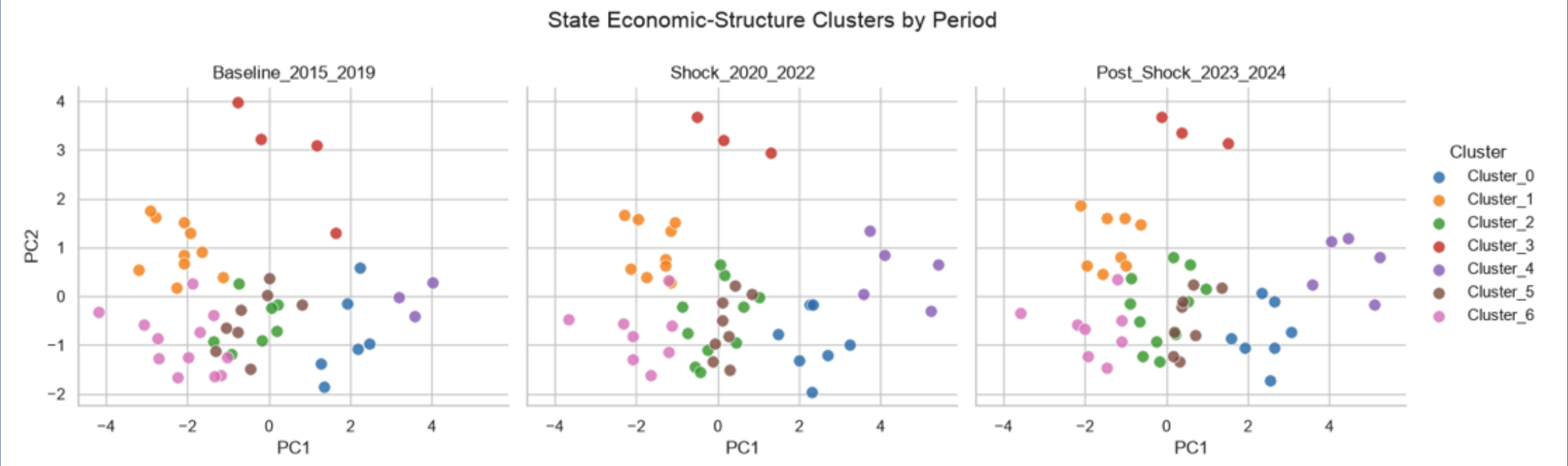
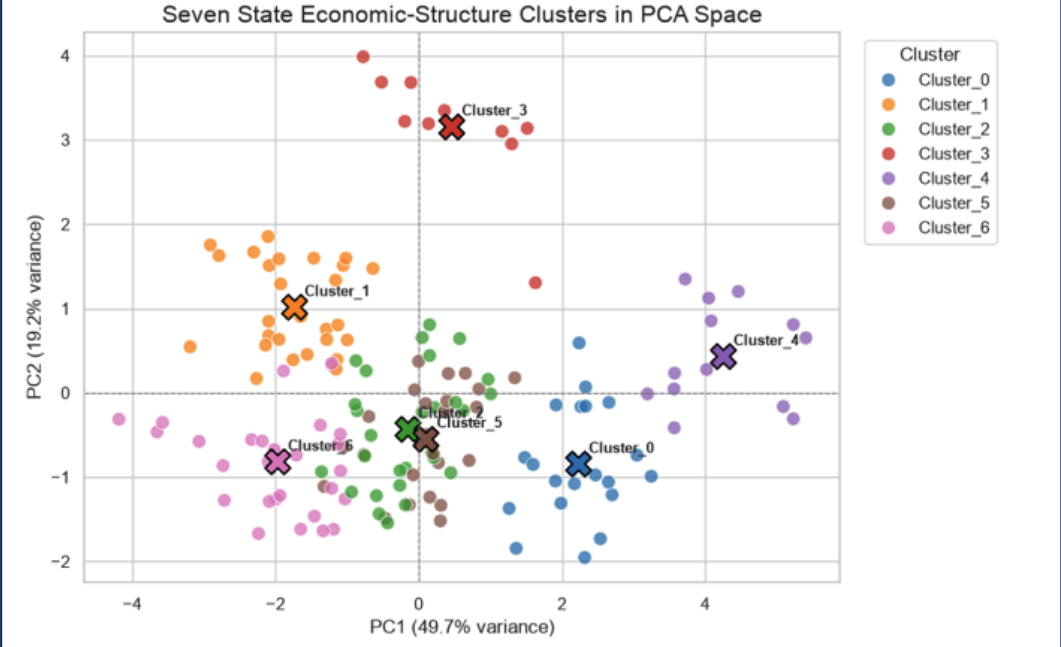
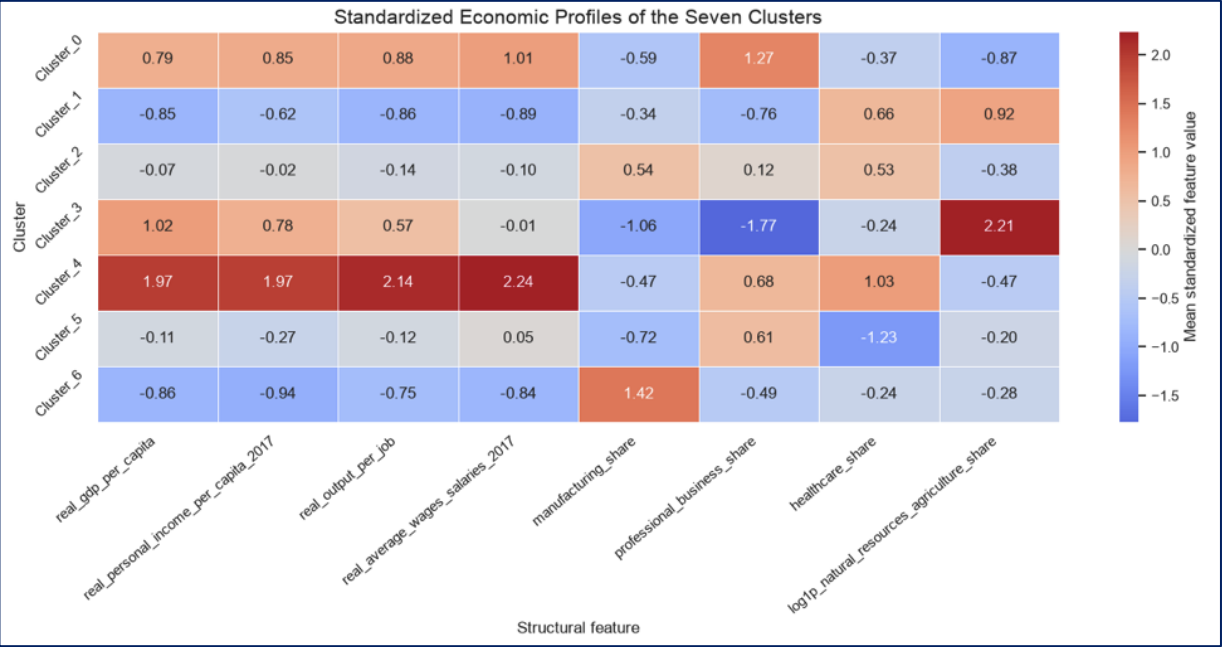
Reason for selecting 7 clusters

Consideration	k = 2	k = 7	k = 8
Silhouette	0.3541	0.3076	0.3186
Calinski–Harabasz	85.2646	73.6397	70.5250
Davies–Bouldin	1.0960	1.0626	1.0561
Mean stability ARI	1.0000	0.9576	0.7826
Minimum stability ARI	1.0000	0.9138	0.6134
Cluster-size range	40–110	10–28	10–26

Selecting k= 7:

- Very strong mean stability: 0.9576
- Strong minimum stability: 0.9138
- Competitive silhouette score: 0.3076
- Competitive Davies–Bouldin score: 1.0626
- Reasonably balanced cluster sizes of 10–28 observations
- Greater economic detail than the two-cluster solution
- Much more stable than the eight-cluster solution

	Cluster_0	Cluster_1	Cluster_2	Cluster_3	Cluster_4	Cluster_5	Cluster_6	Cluster_7
Number_of_Clusters								
2	40	110	0	0	0	0	0	0
3	35	81	34	0	0	0	0	0
4	51	72	10	17	0	0	0	0
5	37	54	31	18	10	0	0	0
6	20	32	25	10	13	50	0	0
7	20	27	28	10	13	24	28	0
8	20	20	10	13	22	24	26	15

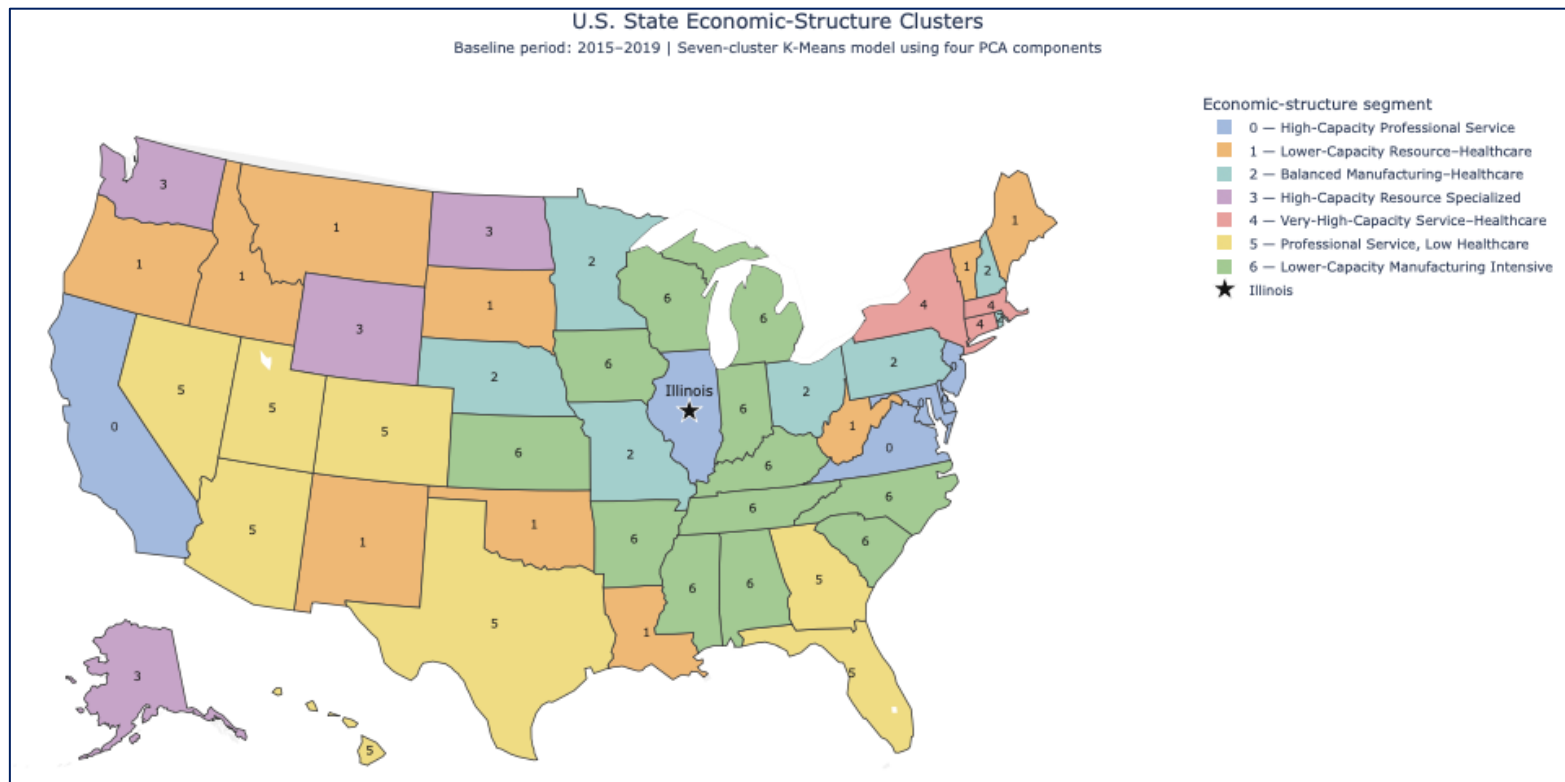


Step 7. Cluster Interpretation

Cluster	Suggested descriptive name	Main evidence
Cluster 0	High-Capacity Professional-Service Economies	Above-average GDP, income, productivity, and wages; professional services +1.273 ; low resource share
Cluster 1	Lower-Capacity Resource–Healthcare Economies	Below-average capacity measures; resource share +0.923 ; healthcare +0.657
Cluster 2	Balanced Manufacturing–Healthcare Economies	Capacity near the national average; manufacturing +0.542 ; healthcare +0.534
Cluster 3	High-Capacity Resource-Specialized Economies	Resource share +2.208 ; GDP +1.023 ; very low professional services and manufacturing
Cluster 4	Very-High-Capacity Service–Healthcare Economies	Extremely high GDP, income, productivity, and wages; healthcare +1.031
Cluster 5	Professional-Service, Low-Healthcare Economies	Professional services +0.608 ; healthcare –1.227 ; other capacity measures near average
Cluster 6	Lower-Capacity Manufacturing-Intensive Economies	Manufacturing +1.417 ; below-average GDP, income, productivity, and wages

Interpretation:

- The baseline map shows that economic-structure clusters.
- Illinois belongs to the High-Capacity Professional-Service cluster with California, Delaware, Maryland, New Jersey, and Virginia.
- The Lower-Capacity Manufacturing-Intensive cluster is concentrated mainly in the Midwest and South. The Very-High-Capacity Service–Healthcare cluster contains Connecticut, Massachusetts, and New York.
- The High-Capacity Resource-Specialized cluster includes Alaska, North Dakota, Washington, and Wyoming. This group is geographically dispersed, showing that cluster membership reflects multivariate economic structure rather than geographic proximity alone.
- The map represents descriptive similarities during the 2015–2019 baseline period and does not imply that geography caused the observed economic profiles



Comparison of U.S. State Economic-Structure Clusters
Consistent cluster definitions and colors across all periods

Baseline: 2015–2019

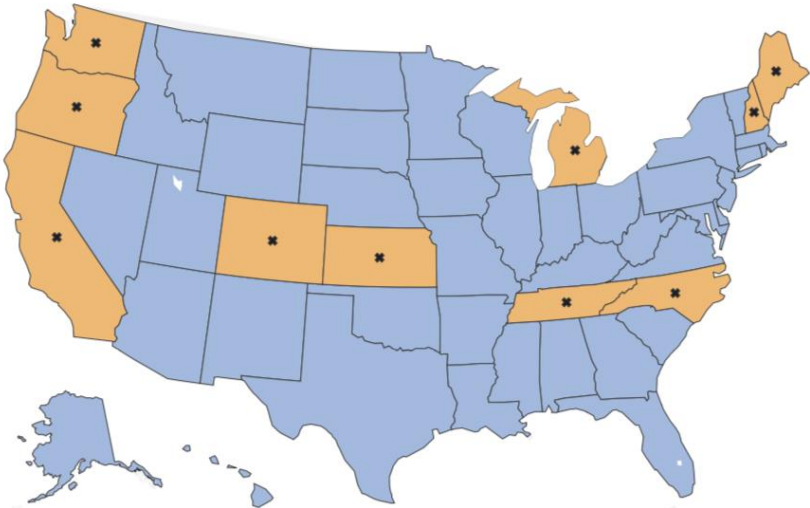
Shock: 2020–2022

Post-shock: 2023–2024



- Economic-structure segment
- 0 — High-Capacity Professional Service
 - 1 — Lower-Capacity Resource–Healthcare
 - 2 — Balanced Manufacturing–Healthcare
 - 3 — High-Capacity Resource Specialized
 - 4 — Very-High-Capacity Service–Healthcare
 - 5 — Professional Service, Low Healthcare
 - 6 — Lower-Capacity Manufacturing Intensive
- 0 — High-Capacity Professional Service
- 1 — Lower-Capacity Resource–Healthcare
- 2 — Balanced Manufacturing–Healthcare
- 3 — High-Capacity Resource Specialized
- 4 — Very-High-Capacity Service–Healthcare
- 5 — Professional Service, Low Healthcare
- 6 — Lower-Capacity Manufacturing Intensive

State Cluster Membership: Baseline vs. Post-Shock
An X identifies a state assigned to a different cluster in 2023–2024



- Baseline-to-post comparison
- Same cluster
 - Changed cluster
 - Changed-cluster marker

	Status	Number_of_States	Percent_of_States
0	Same cluster	40	80.0000
1	Changed cluster	10	20.0000

Step 8. Cluster path changes Interpretation from three periods

Structural status definitions

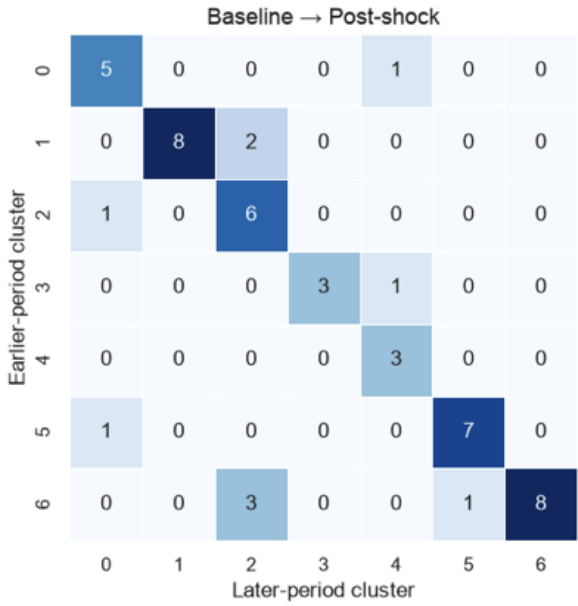
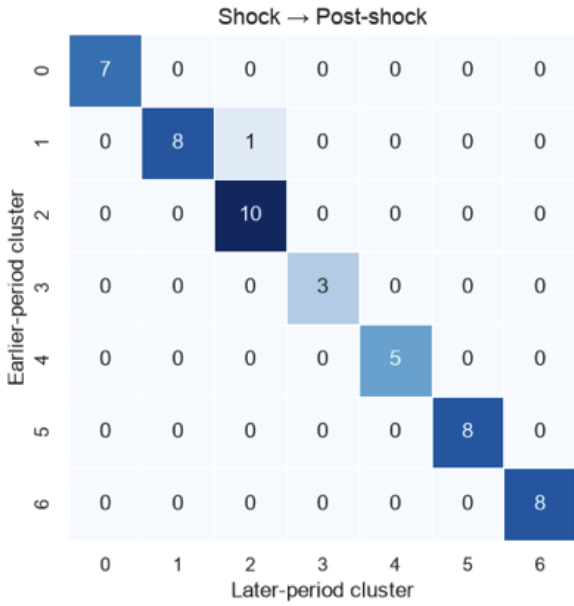
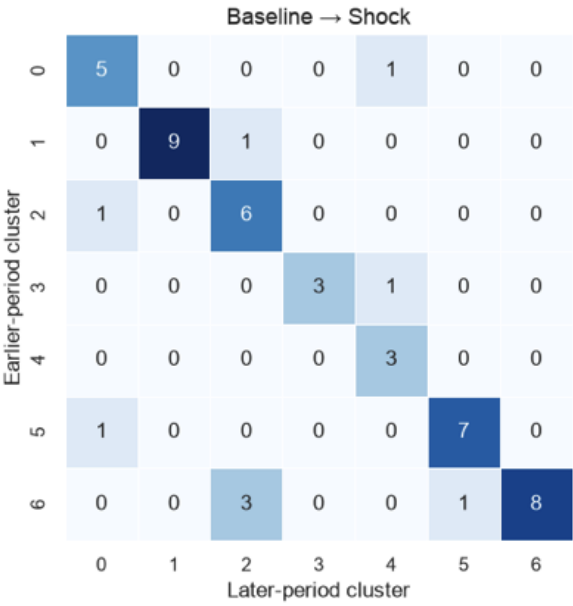
Example_Path		Structural_Status	Interpretation
0	A → A → A	Structurally stable	The state remained in the same structural cluster across all three periods.
1	A → B → A	Temporary shock-period change	The state changed during the shock period and returned to its baseline cluster.
2	A → B → B	Possible shock-origin persistent shift	The state changed during the shock period and remained in that cluster afterward.
3	A → A → B	Possible post-shock shift	The state remained stable during the shock but changed during the post-shock period.
4	A → B → C	Complex or continuing transition	The state followed three different or otherwise more complicated cluster positions.

Structural status observations

state	Cluster_Path	Structural_Status
0	Alabama 6 → 6 → 6	Structurally stable
1	Alaska 3 → 3 → 3	Structurally stable
2	Arizona 5 → 5 → 5	Structurally stable
3	Arkansas 6 → 6 → 6	Structurally stable
6	Connecticut 4 → 4 → 4	Structurally stable
7	Delaware 0 → 0 → 0	Structurally stable
8	Florida 5 → 5 → 5	Structurally stable
9	Georgia 5 → 5 → 5	Structurally stable
10	Hawaii 5 → 5 → 5	Structurally stable
11	Idaho 1 → 1 → 1	Structurally stable
12	Illinois 0 → 0 → 0	Structurally stable
13	Indiana 6 → 6 → 6	Structurally stable
14	Iowa 6 → 6 → 6	Structurally stable
16	Kentucky 6 → 6 → 6	Structurally stable
17	Louisiana 1 → 1 → 1	Structurally stable
19	Maryland 0 → 0 → 0	Structurally stable
20	Massachusetts 4 → 4 → 4	Structurally stable
22	Minnesota 2 → 2 → 2	Structurally stable
23	Mississippi 6 → 6 → 6	Structurally stable
24	Missouri 2 → 2 → 2	Structurally stable
25	Montana 1 → 1 → 1	Structurally stable
26	Nebraska 2 → 2 → 2	Structurally stable
27	Nevada 5 → 5 → 5	Structurally stable
29	New Jersey 0 → 0 → 0	Structurally stable
30	New Mexico 1 → 1 → 1	Structurally stable
31	New York 4 → 4 → 4	Structurally stable
33	North Dakota 3 → 3 → 3	Structurally stable
34	Ohio 2 → 2 → 2	Structurally stable
35	Oklahoma 1 → 1 → 1	Structurally stable
37	Pennsylvania 2 → 2 → 2	Structurally stable
38	Rhode Island 2 → 2 → 2	Structurally stable
39	South Carolina 6 → 6 → 6	Structurally stable
40	South Dakota 1 → 1 → 1	Structurally stable
42	Texas 5 → 5 → 5	Structurally stable
43	Utah 5 → 5 → 5	Structurally stable
44	Vermont 1 → 1 → 1	Structurally stable
45	Virginia 0 → 0 → 0	Structurally stable
47	West Virginia 1 → 1 → 1	Structurally stable
48	Wisconsin 6 → 6 → 6	Structurally stable
49	Wyoming 3 → 3 → 3	Structurally stable
4	California 0 → 4 → 4	Possible shock-origin persistent shift
5	Colorado 5 → 0 → 0	Possible shock-origin persistent shift
15	Kansas 6 → 2 → 2	Possible shock-origin persistent shift
21	Michigan 6 → 2 → 2	Possible shock-origin persistent shift
28	New Hampshire 2 → 0 → 0	Possible shock-origin persistent shift
32	North Carolina 6 → 5 → 5	Possible shock-origin persistent shift
36	Oregon 1 → 2 → 2	Possible shock-origin persistent shift
41	Tennessee 6 → 2 → 2	Possible shock-origin persistent shift
46	Washington 3 → 4 → 4	Possible shock-origin persistent shift
18	Maine 1 → 1 → 2	Possible post-shock shift

Structural_Status		Number_of_States	States
0	Structurally stable	40	Alabama, Alaska, Arizona, Arkansas, Connecticut, Delaware, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kentucky, Louisiana, Maryland, Massachusetts, Minnesota, Mississippi, Missouri, Montana, Nebraska, Nevada, New Jersey, New Mexico, New York, North Dakota, Ohio, Oklahoma, Pennsylvania, Rhode Island, South Carolina, South Dakota, Texas, Utah, Vermont, Virginia, West Virginia, Wisconsin, Wyoming
1	Temporary shock-period change	0	
2	Possible shock-origin persistent shift	9	California, Colorado, Kansas, Michigan, New Hampshire, North Carolina, Oregon, Tennessee, Washington
3	Possible post-shock shift	1	Maine
4	Complex or continuing transition	0	

State Economic-Structure Cluster Transitions



Growth by status and periods

		real_gdp_per_capita_growth	real_income_per_capita_growth	total_employment_growth	real_average_wage_growth
Structural_Status	period				
Structurally stable	Baseline_2015_2019	1.0910	1.3300	1.0210	0.6010
	Shock_2020_2022	1.2690	1.7070	1.3230	0.9750
	Post_Shock_2023_2024	2.1870	1.3200	1.8390	0.4290
Possible shock-origin persistent shift	Baseline_2015_2019	2.1050	2.0670	1.5380	1.2190
	Shock_2020_2022	2.1470	1.9030	1.3070	1.6420
	Post_Shock_2023_2024	1.8110	1.4940	1.6930	0.5770
Possible post-shock shift	Baseline_2015_2019	2.0530	1.9250	0.7820	0.9940
	Shock_2020_2022	2.1350	2.5980	1.0450	2.6920
	Post_Shock_2023_2024	2.6700	2.0620	2.0240	0.5860

Interpretation:

- State economic and industry-employment structures were predominantly stable across the baseline, shock, and post-shock periods. Forty states, representing 80% of the sample, remained in the same structural cluster throughout all three periods.
- Nine states—California, Colorado, Kansas, Michigan, New Hampshire, North Carolina, Oregon, Tennessee, and Washington—changed clusters between the 2015–2019 baseline and the 2020–2022 shock period. All nine remained in their new clusters during 2023–2024. These pathways are classified **as possible shock-origin persistent shifts**.
- **Maine** was the only state that remained in its baseline cluster during 2020–2022 but changed clusters during 2023–2024. It is classified **as a possible post-shock shift**.
- No state exhibited a temporary shock-period change by moving to a different cluster and then returning to its baseline cluster. No state followed a complex three-cluster transition.
- The transition analysis indicates that interstate structural similarities were largely stable. Most changes occurred between the baseline and shock periods, and those changes generally persisted during the available post-shock period.
- These results are descriptive. A persistent cluster change means that a state became relatively closer to a different multivariate cluster centroid. **It does not prove that the economic shock caused a permanent structural transformation.** The 2023–2024 post-shock window is also too short to establish long-term permanence.

Step 9. Illinois Economic Peer Analysis

- Illinois's closest economic peers are identified separately during the baseline, shock, and post-shock periods.
- Similarity is measured using Euclidean distance across PC1-PC4. These four components preserve 91.82% of the standardized structural information.
- A smaller distance indicates that another state has a structural profile more similar to Illinois.

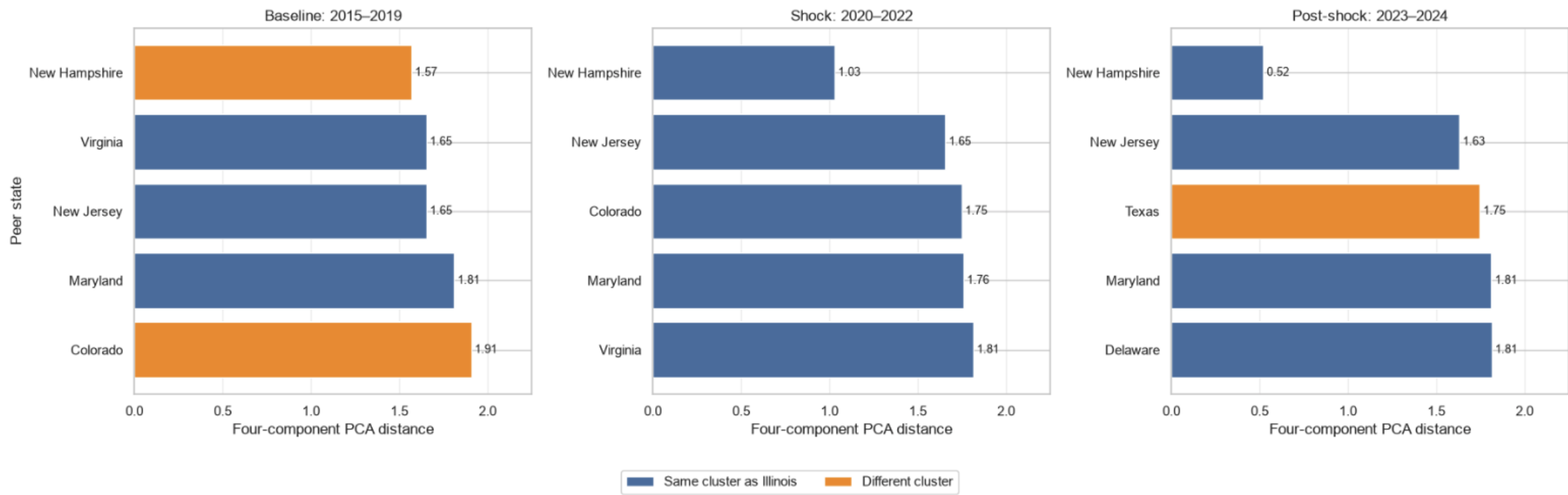
Input setting

Target state	Illinois
Distance dimensions	PC1, PC2, PC3, PC4
Peers retained per period	150

For state j , the distance from Illinois is:

$$d_{IL,j} = \sqrt{(PC1_{IL} - PC1_j)^2 + (PC2_{IL} - PC2_j)^2 + (PC3_{IL} - PC3_j)^2 + (PC4_{IL} - PC4_j)^2}$$

Illinois's Closest Economic Peers by Period

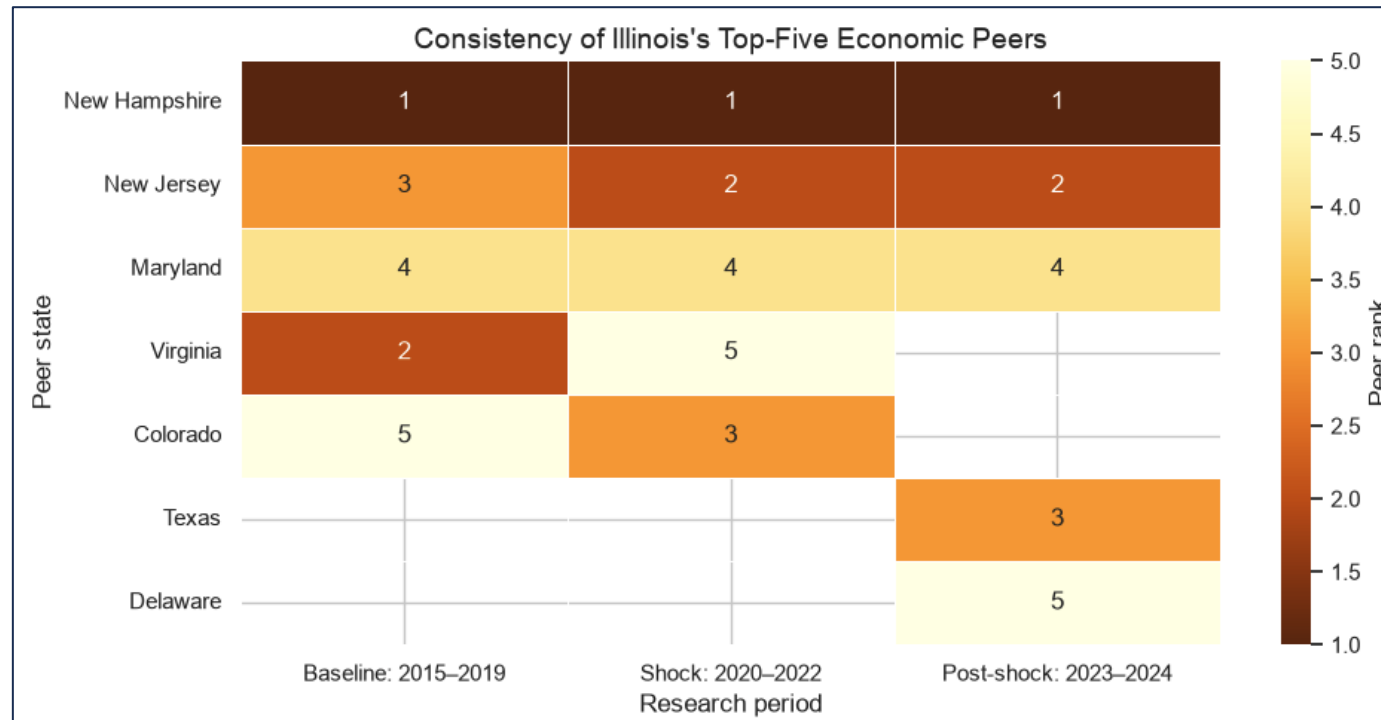


	Period_Label	Closest_Peer	Closest_Distance	Top_Five_Peers
0	Baseline: 2015–2019	New Hampshire	1.5679	New Hampshire, Virginia, New Jersey, Maryland, Colorado
1	Shock: 2020–2022	New Hampshire	1.0282	New Hampshire, New Jersey, Colorado, Maryland, Virginia
2	Post-shock: 2023–2024	New Hampshire	0.5193	New Hampshire, New Jersey, Texas, Maryland, Delaware

	Comparison	Shared_Top_Five_Peers	Shared_States	Jaccard_Similarity
0	Baseline → Shock	5	Colorado, Maryland, New Hampshire, New Jersey, Virginia	1.0000
1	Shock → Post-shock	3	Maryland, New Hampshire, New Jersey	0.4286
2	Baseline → Post-shock	3	Maryland, New Hampshire, New Jersey	0.4286

Interpretation:

- **New Hampshire** was Illinois's closest economic peer during all three research periods. Its PCA distance from Illinois declined from **1.5679** during the 2015–2019 baseline period to **1.0282** during 2020–2022 and **0.5193** during 2023–2024. This pattern indicates increasing multivariate structural similarity.
- **New Jersey** and **Maryland** also appeared among Illinois's five closest peers during all three periods. Together with New Hampshire, they form the most consistent core of Illinois's economic peer group.
- The baseline and shock-period top-five peer lists contained the same five states: Colorado, Maryland, New Hampshire, New Jersey, and Virginia. Their Jaccard similarity was 1.0000, although their rankings changed.
- The post-shock peer group changed more substantially. **New Hampshire, New Jersey, and Maryland remained**, while Texas and Delaware replaced Colorado and Virginia. The baseline-to-post-shock and shock-to-post-shock Jaccard similarities were both **0.4286**.
- **Illinois** remained in Cluster 0—the **High-Capacity Professional-Service Economies cluster**—during all three periods. Nevertheless, the change in its closest peers shows that stable broad cluster membership does not necessarily imply an unchanged position within the cluster.
- The PCA distances are relative multivariate measures without natural economic units. A smaller distance indicates greater similarity across PC1–PC4, but it does not mean that two states have identical economies or establish a causal relationship.



Findings and Conclusions

□ The findings that indicate structural similarities among state change during shock and post-shock periods:

- **41 states (82%)** remained in their baseline cluster during 2020–2022.
- **9 states (18%)** changed clusters during 2020–2022.
- All nine remained in their new cluster during 2023–2024.
- Only **Maine** changed between the shock and post-shock periods.
- Overall, **40 states (80%)** occupied the same cluster in the baseline and post-shock periods.
- **10 states (20%)** occupied a different post-shock cluster.

□ The states experienced a possible shock-origin persistent shifts across the three periods:

Table 1

Structural status	States	Percentage
Structurally stable	40	80%
Temporary shock-period change	0	0%
Possible shock-origin persistent shift	9	18%
Possible post-shock shift	1	2%
Complex or continuing transition	0	0%

Table 2

State	Path	Descriptive movement
California	0 → 4 → 4	High-capacity professional service → very-high-capacity service–healthcare
Colorado	5 → 0 → 0	Professional service, low healthcare → high-capacity professional service
Kansas	6 → 2 → 2	Manufacturing-intensive, lower capacity → balanced manufacturing–healthcare
Michigan	6 → 2 → 2	Manufacturing-intensive, lower capacity → balanced manufacturing–healthcare
New Hampshire	2 → 0 → 0	Balanced manufacturing–healthcare → high-capacity professional service
North Carolina	6 → 5 → 5	Manufacturing-intensive, lower capacity → professional service, low healthcare
Oregon	1 → 2 → 2	Resource–healthcare, lower capacity → balanced manufacturing–healthcare
Tennessee	6 → 2 → 2	Manufacturing-intensive, lower capacity → balanced manufacturing–healthcare
Washington	3 → 4 → 4	Resource-specialized, high capacity → very-high-capacity service–healthcare

Hierarchical Clustering Evaluation

Method 1. Pooled evaluation

Description: Use all 150 state-period observations to compare hierarchical clustering with the existing seven-cluster K-means model.

Method 2. Period-specific evaluation

Description: Apply hierarchical clustering separately to the baseline, shock, and post-shock periods.

K-MEANS reference Scores

Model		Linkage	K	Silhouette	Calinski_Harabasz	Davies_Bouldin
0	K-means	Not applicable	7	0.3076	73.6397	1.0626

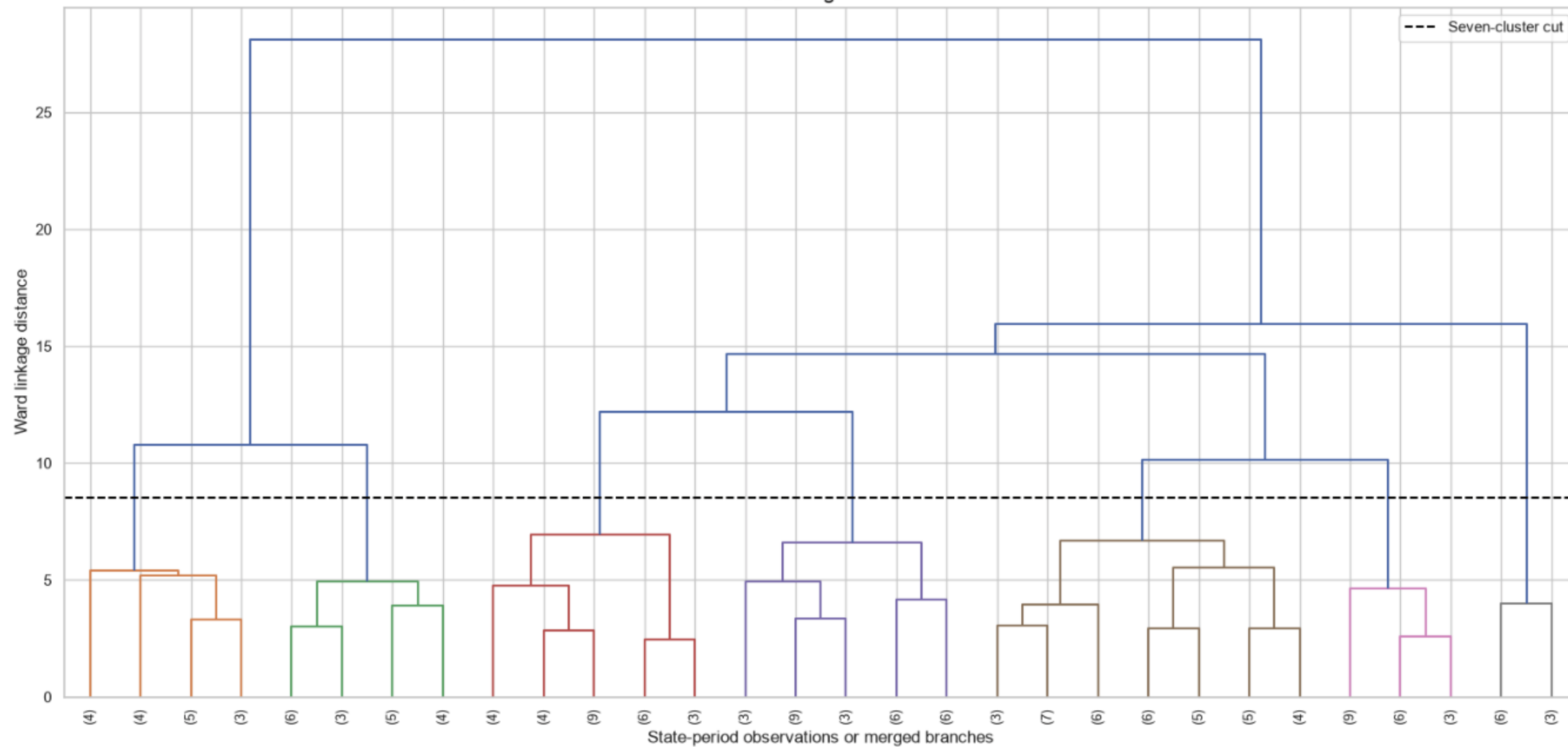
Pooled hierarchical model results

	Model	Linkage	K	Silhouette	Calinski_Harabasz	Davies_Bouldin	Smallest_Cluster	Largest_Cluster
0	Hierarchical	Ward	2	0.3607	82.6005	1.0212	34	116
1	Hierarchical	Ward	3	0.3576	66.1940	0.8972	9	107
2	Hierarchical	Ward	4	0.2497	64.8665	1.2986	9	54
3	Hierarchical	Ward	5	0.2720	64.0588	1.1329	9	54
4	Hierarchical	Ward	6	0.2610	64.4607	1.1389	9	54
5	Hierarchical	Ward	7	0.2841	67.0326	1.0840	9	36
6	Hierarchical	Ward	8	0.2957	64.0575	1.0348	9	36
7	Hierarchical	Complete	2	0.3207	75.0807	1.1274	38	112
8	Hierarchical	Complete	3	0.3324	61.3141	0.9557	9	103
9	Hierarchical	Complete	4	0.2664	52.8615	1.0281	9	103
10	Hierarchical	Complete	5	0.2210	56.7269	1.2891	9	65
11	Hierarchical	Complete	6	0.2543	62.4277	1.2437	9	38
12	Hierarchical	Complete	7	0.2583	61.9303	1.1771	9	34
13	Hierarchical	Complete	8	0.2473	59.6416	1.1234	5	34
14	Hierarchical	Average	2	0.3721	48.3095	0.8239	14	136
15	Hierarchical	Average	3	0.3137	41.4042	0.8221	9	127
16	Hierarchical	Average	4	0.2413	32.4707	0.7713	3	124
17	Hierarchical	Average	5	0.2732	54.0912	0.9388	3	77
18	Hierarchical	Average	6	0.2290	47.0626	0.9274	3	73
19	Hierarchical	Average	7	0.2539	48.9736	0.9075	3	60
20	Hierarchical	Average	8	0.2632	54.3555	0.9512	3	60
21	Hierarchical	Single	2	0.2763	17.7962	0.9446	9	141
22	Hierarchical	Single	3	0.1973	13.0922	0.8513	3	138
23	Hierarchical	Single	4	0.1423	13.9644	0.7804	3	134
24	Hierarchical	Single	5	0.2212	25.6905	0.7543	3	125
25	Hierarchical	Single	6	0.2041	20.8656	0.7056	3	125
26	Hierarchical	Single	7	0.1146	17.8304	0.7052	1	124
27	Hierarchical	Single	8	0.0914	16.6511	0.7371	1	121

Conclusion: Best pooled hierarchical model

Linkage	Ward
Number of clusters	3
Silhouette	0.3576
Calinski-Harabasz	66.194
Davies-Bouldin	0.8972
Smallest cluster	9

Pooled Hierarchical Structure: Ward Linkage Across 150 State-Period Observations



Pooled cophenetic correlation: 0.6489

Select the best hierarchical K-MEANS (k= 7) model

	Model	Linkage	K	Silhouette	Calinski_Harabasz	Davies_Bouldin	Smallest_Cluster	Largest_Cluster	Eligible_for_Selection	Silhouette_Rank	Calinski_Harabasz_Rank	Davies_Bouldin_Rank	Mean_Metric_Rank
0	Hierarchical	Ward	7	0.2841	67.0326	1.0840	9	36	True	1.0000	1.0000	2.0000	1.3333
1	Hierarchical	Complete	7	0.2583	61.9303	1.1771	9	34	True	2.0000	2.0000	3.0000	2.3333
2	Hierarchical	Average	7	0.2539	48.9736	0.9075	3	60	True	3.0000	3.0000	1.0000	2.3333

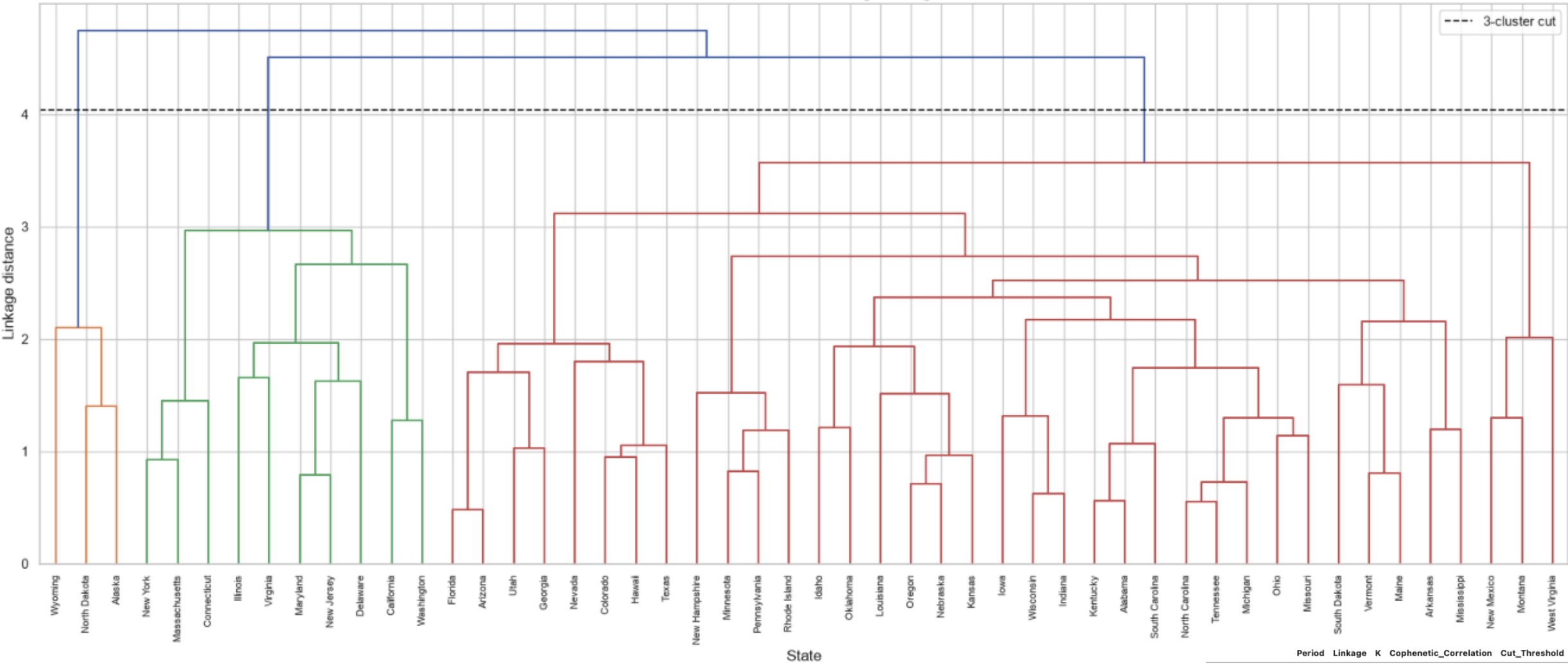
Compare K-Means with hierarchical k= 7

	Model	Linkage	K	Silhouette	Calinski_Harabasz	Davies_Bouldin
0	K-means	Not applicable	7	0.3076	73.6397	1.0626
1	Hierarchical	Ward	7	0.2841	67.0326	1.0840
Adjusted Rand Index between K-means and hierarchical clustering: 0.7458						

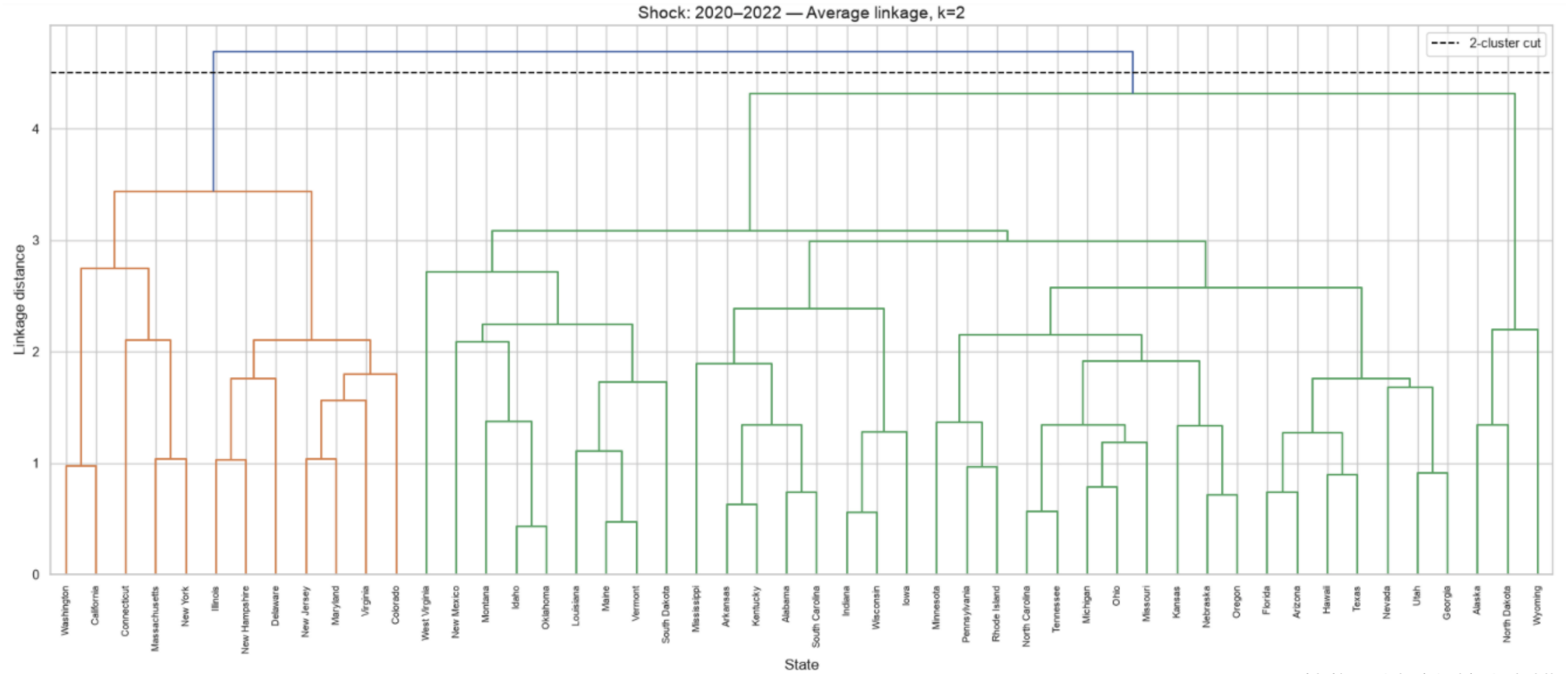
Interpretations:

- Higher silhouette: better-defined clusters.
- Higher Calinski-Harabasz: stronger separation.
- Lower Davies-Bouldin: lower overlap.
- Higher ARI: greater agreement between the two algorithms.
- ARI measures agreement, not which model is better.

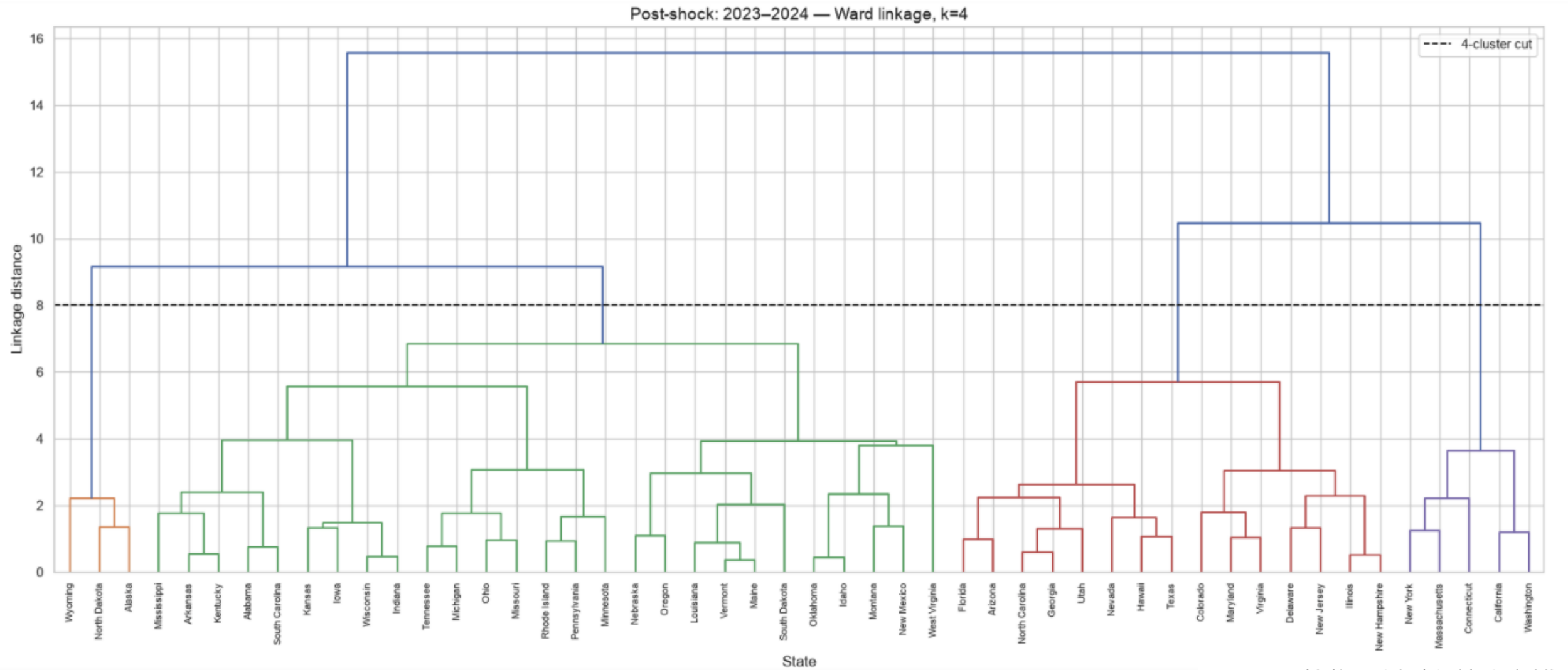
Baseline: 2015–2019 — Average linkage, k=3



	Period	Linkage	K	Cophenetic_Correlation	Cut_Threshold
0	Baseline_2015_2019	Average	3	0.7441	4.0354
1	Shock_2020_2022	Average	2	0.7014	4.5027
2	Post_Shock_2023_2024	Ward	4	0.5779	8.0009



	Period	Linkage	K	Cophenetic_Correlation	Cut_Threshold
0	Baseline_2015_2019	Average	3	0.7441	4.0354
1	Shock_2020_2022	Average	2	0.7014	4.5027
2	Post_Shock_2023_2024	Ward	4	0.5779	8.0009



	Period	Linkage	K	Cophenetic_Correlation	Cut_Threshold
0	Baseline_2015_2019	Average	3	0.7441	4.0354
1	Shock_2020_2022	Average	2	0.7014	4.5027
2	Post_Shock_2023_2024	Ward	4	0.5779	8.0009

Rank models within each period

Period	Best linkage	(k)	Silhouette	Cluster sizes	Interpretation
Baseline	Average	3	0.3522	3–37	One dominant cluster
Shock	Average	2	0.3628	12–38	Primarily a two-group division
Post-shock	Ward	4	0.3234	3–27	More differentiated structure

Comparison table

Metric	K-means	Hierarchical Ward	Better model
Silhouette	0.3076	0.2841	K-means
Calinski–Harabasz	73.6397	67.0326	K-means
Davies–Bouldin	1.0626	1.0840	K-means
Number of clusters	7	7	Same

Conclusion:

The ***seven-cluster K-means model*** outperformed the comparable ***seven-cluster hierarchical model*** across all three internal evaluation metrics. However, the Adjusted Rand Index of 0.7458 indicates substantial agreement between the two algorithms. Hierarchical clustering therefore supports the broad structural patterns identified by K-means, although it favors fewer and more unevenly sized clusters. **K-means remains the primary model**, while hierarchical clustering serves as a robustness and visualization method

Limitations

This Project Cannot Conclude:

- That the pandemic **caused** a particular cluster change
- That one cluster is economically "**better**"
- That cluster membership predicts future growth
- That correlated features have causal effects

Reference Sources

Data

- ❖ U.S. Bureau of Economic Analysis. (2026). *GDP by state*. <https://www.bea.gov/data/gdp/gdp-state>
- ❖ U.S. Bureau of Economic Analysis. (2026). *Personal income by state*. <https://www.bea.gov/data/income-saving/personal-income-by-state>
- ❖ U.S. Bureau of Labor Statistics. (2026). *Quarterly Census of Employment and Wages: Overview*. <https://www.bls.gov/cew/overview.htm>
- ❖ U.S. Bureau of Labor Statistics. (2026). *Local Area Unemployment Statistics*. <https://www.bls.gov/lau/>
- ❖ U.S. Census Bureau. (2026). *American Community Survey data via API*. <https://www.census.gov/programs-surveys/acs/data/data-via-api.html>
- ❖ The QCEW program provides employment and wage data by state and detailed industry, while LAUS provides state and local labor-force measures. [BLS QCEW](#), [BLS LAUS](#)

Python Libraries

Python library	Project use
pandas	Loading, cleaning, merging, reshaping, aggregating, and organizing state-level data
NumPy	Numerical calculations, arrays, and distance-related calculations
scikit-learn	StandardScaler, PCA, clustering, silhouette scores, and other evaluation metrics
SciPy	Hierarchical clustering, linkage calculations, dendrograms, and distance functions
Matplotlib	General charts and analytical visualizations
Seaborn	Correlation heatmaps, distributions, and statistical charts
Jupyter Notebook	Developing, documenting, executing, and validating the Python workflow